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Silha et al.

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(54) **BATTERY PACK AND CHARGER SYSTEM**

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This patent is subject to a terminal disclaimer.

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(63) Continuation of application No. 18/308,350, filed on Apr. 27, 2023, now Pat. No. 12,015,130, which is a (Continued)

(51) **Int. Cl.**
H01M 10/46 (2006.01)
H01M 10/44 (2006.01)
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CPC **H01M 10/441** (2013.01); **H01M 10/6563** (2015.04); **H01M 50/20** (2021.01); **H02J 7/0013** (2013.01); **H02J 7/0044** (2013.01)

(58) **Field of Classification Search**

CPC H02J 7/0047; H02J 7/0013; H02J 7/0044; H02J 7/0042; H02J 7/00047; H01M 10/441; H01M 10/6563; H01M 50/204
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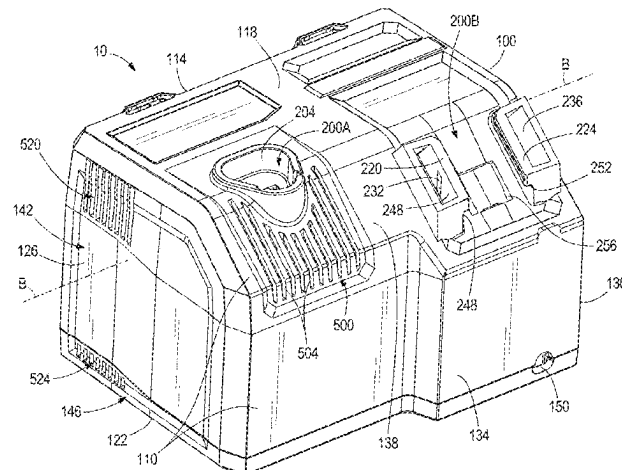
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(57) **ABSTRACT**

A charger includes a housing with an interface positioned in a front wall and configured to engage a battery pack. The interface includes charging terminals positioned between a first rail and a second rail, a first groove positioned between the first rail and a wall of the housing, and a second groove positioned between the second rail and the wall of the housing. The interface is in communication with an interior of the housing. A fan is coupled within the housing adjacent the interface, and an air passage member includes a hollow body that has a first end coupled to the fan and a second end spaced apart from the first end extending through another wall of the housing. The fan is operable to suck an air flow

(Continued)



into the housing from outside the housing and guide the air flow through the air passage member to the interface.

19 Claims, 24 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/110,456, filed on Dec. 3, 2020, now Pat. No. 11,670,808.

- (60) Provisional application No. 62/942,889, filed on Dec. 3, 2019.

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H02J 7/00 (2006.01)

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USPC 320/107, 114, 116, 150, 153
See application file for complete search history.

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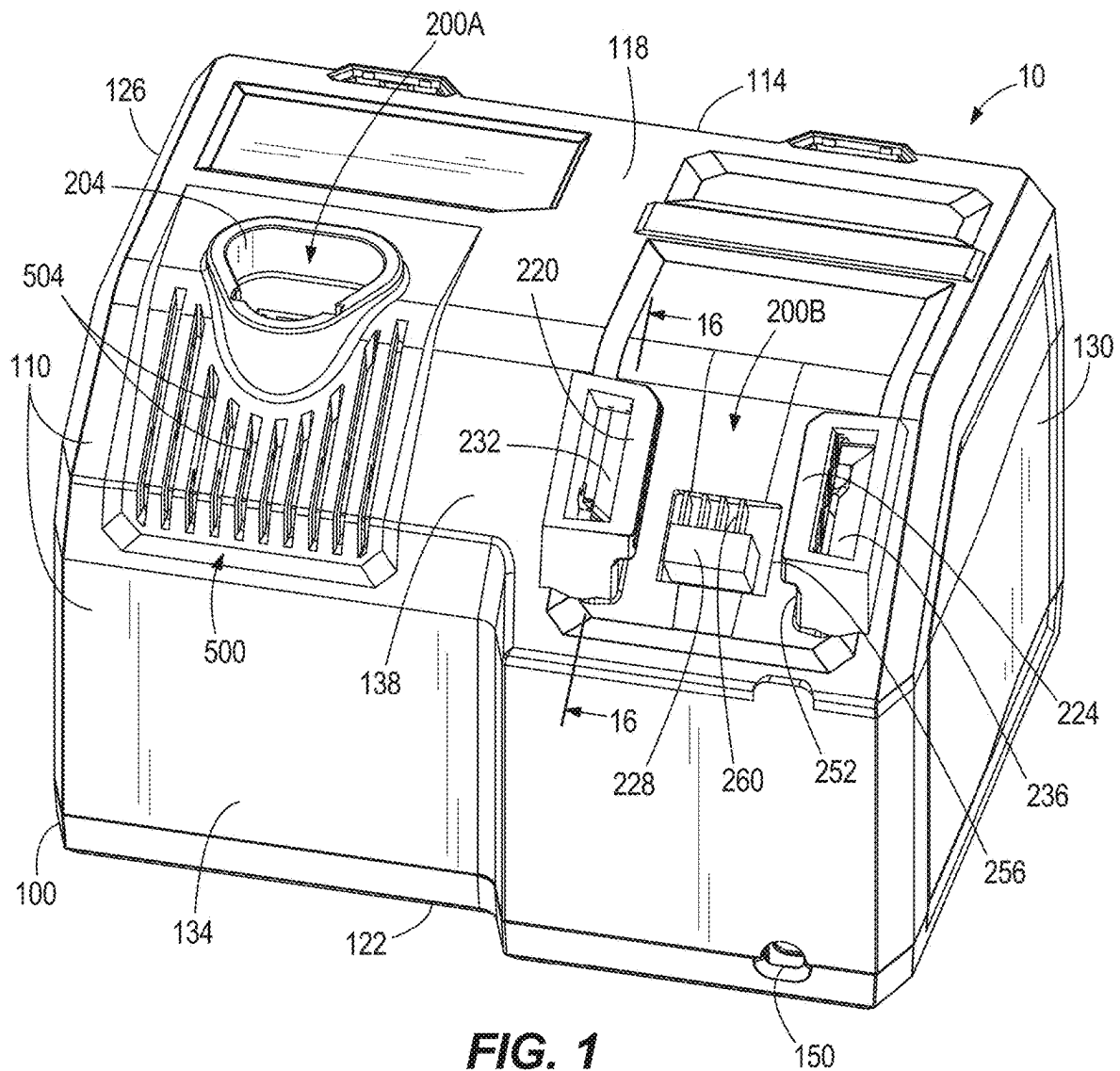


FIG. 1

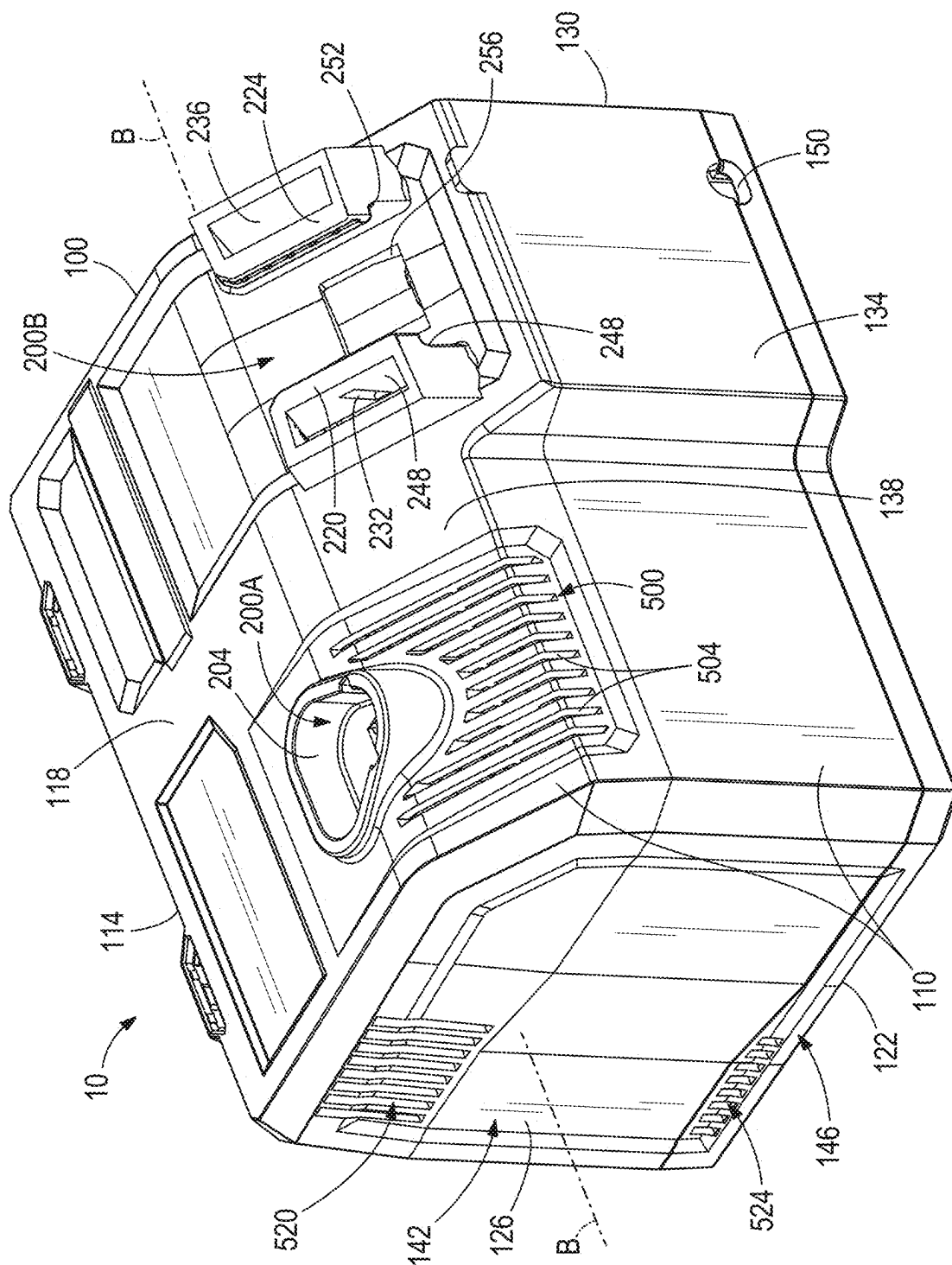


FIG. 2

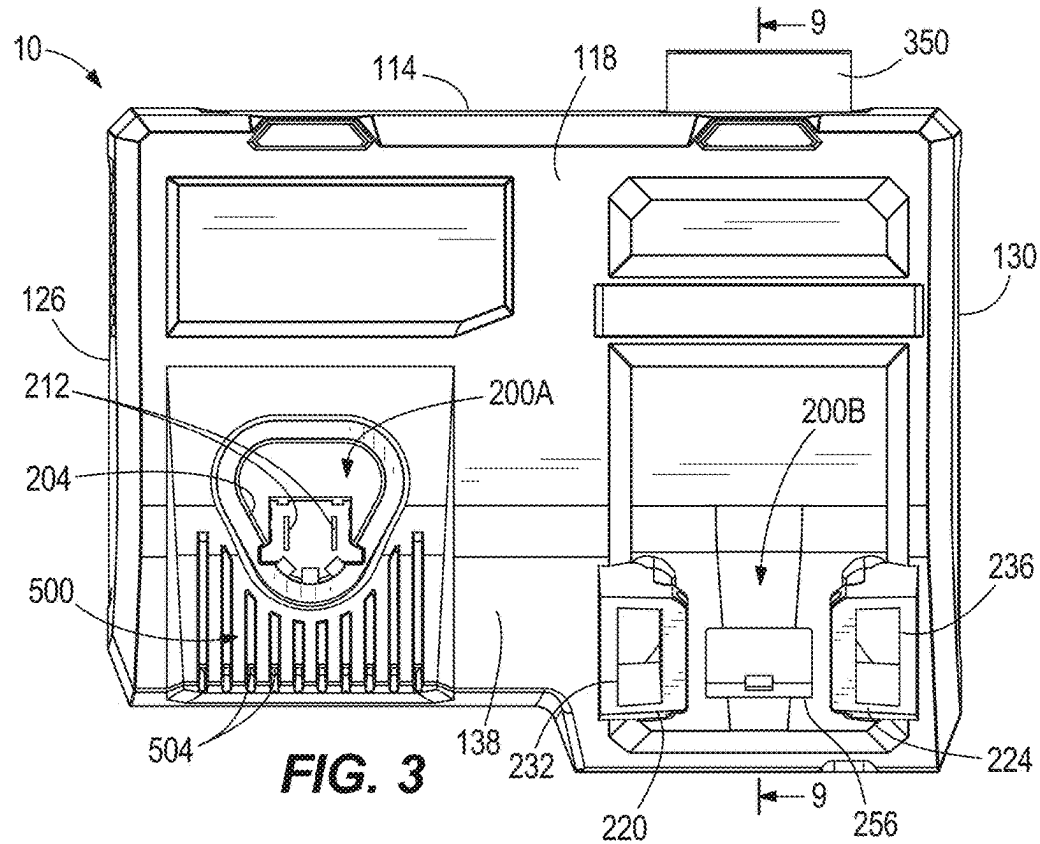


FIG. 3

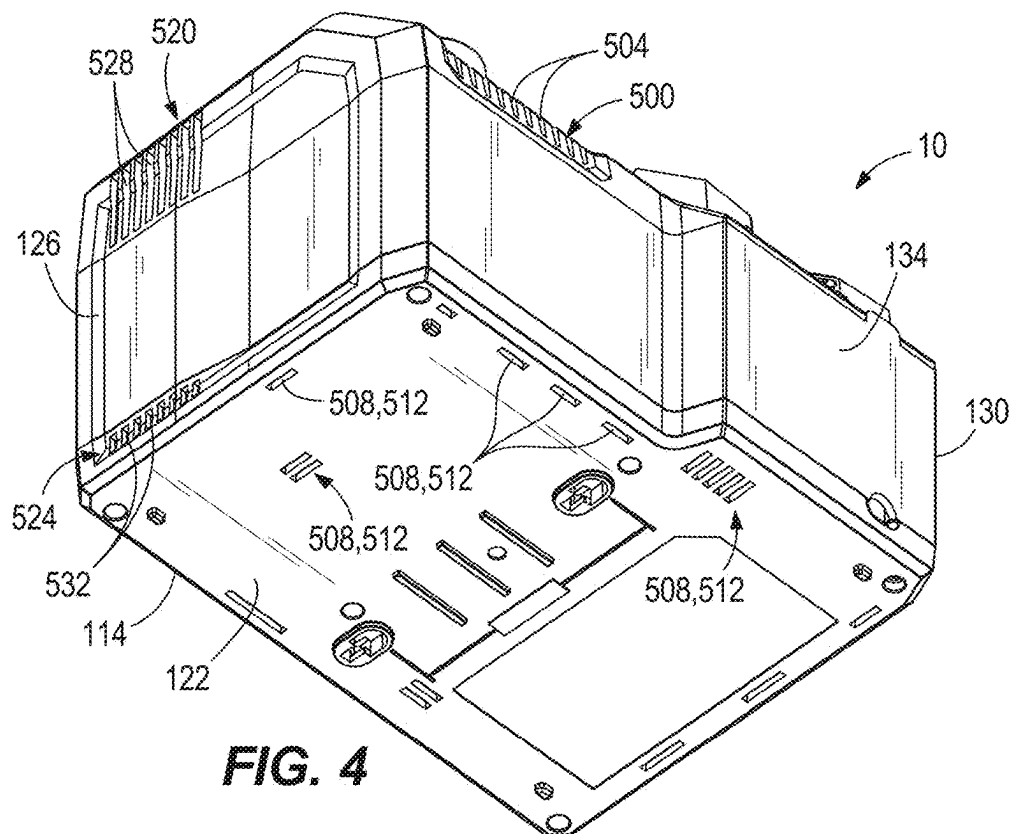
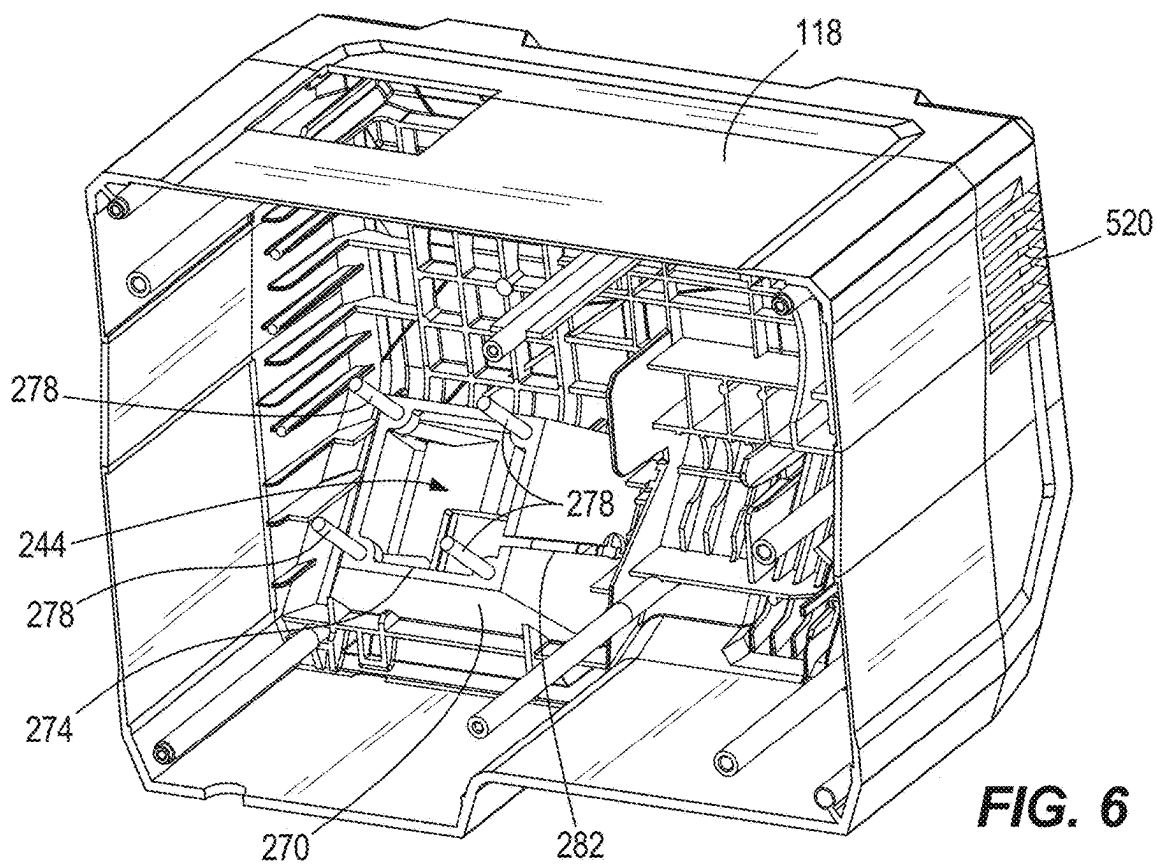
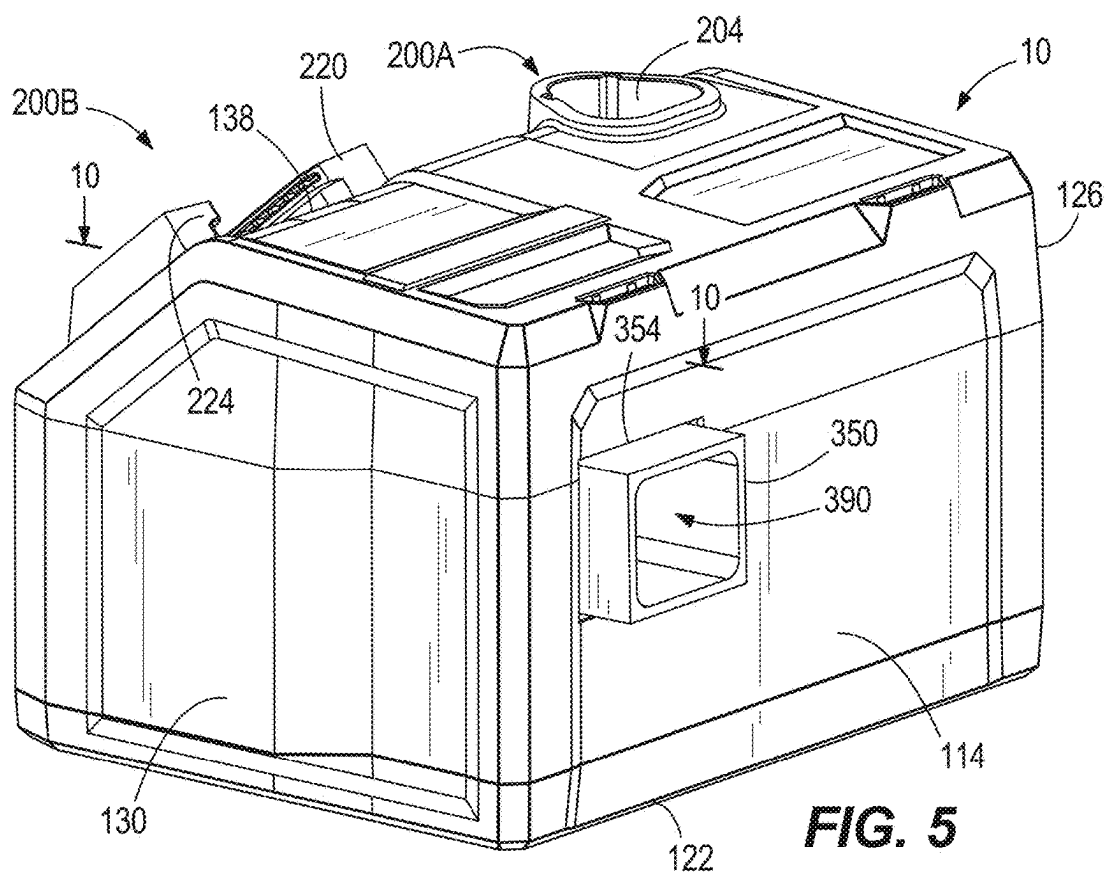


FIG. 4



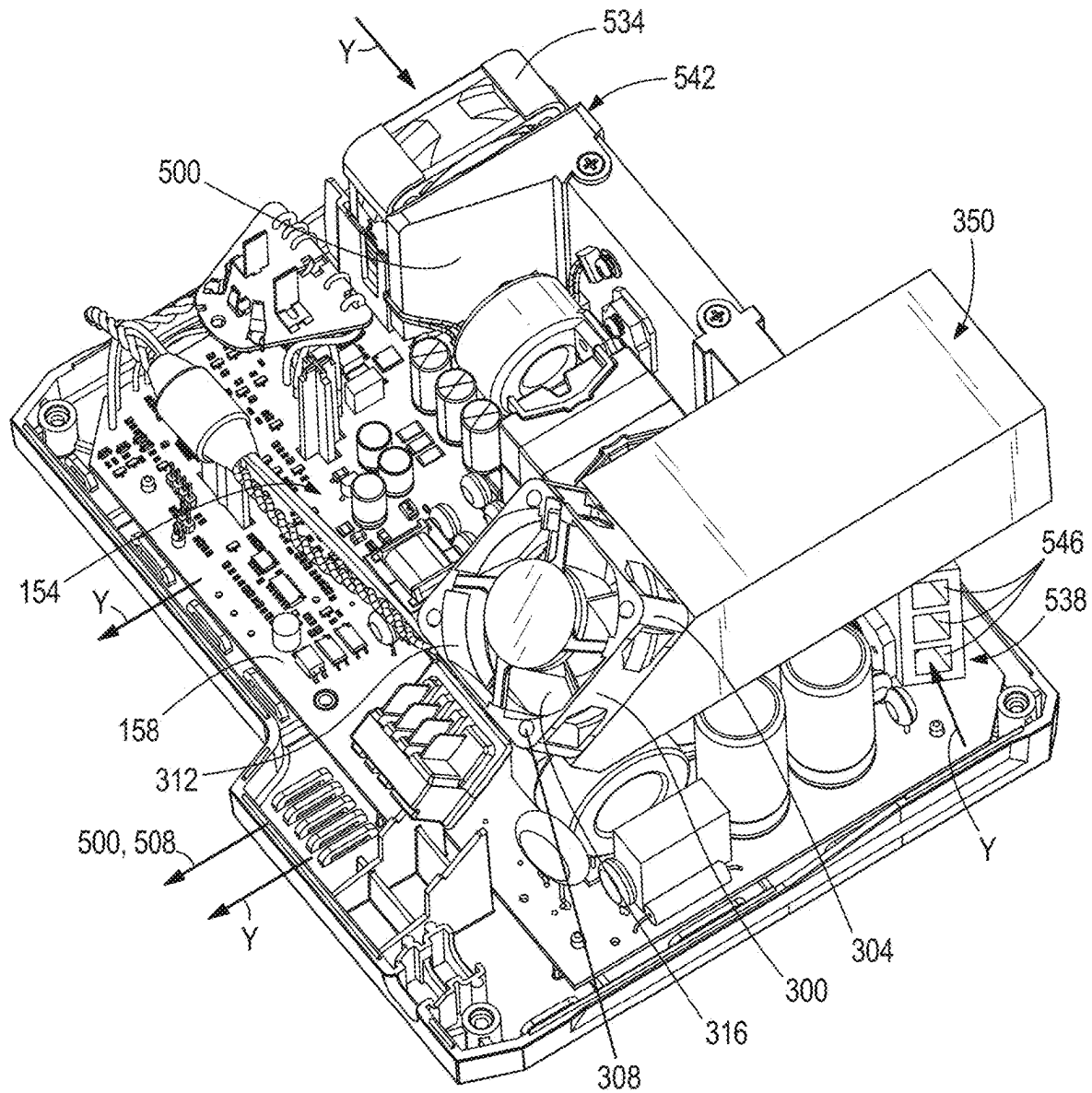
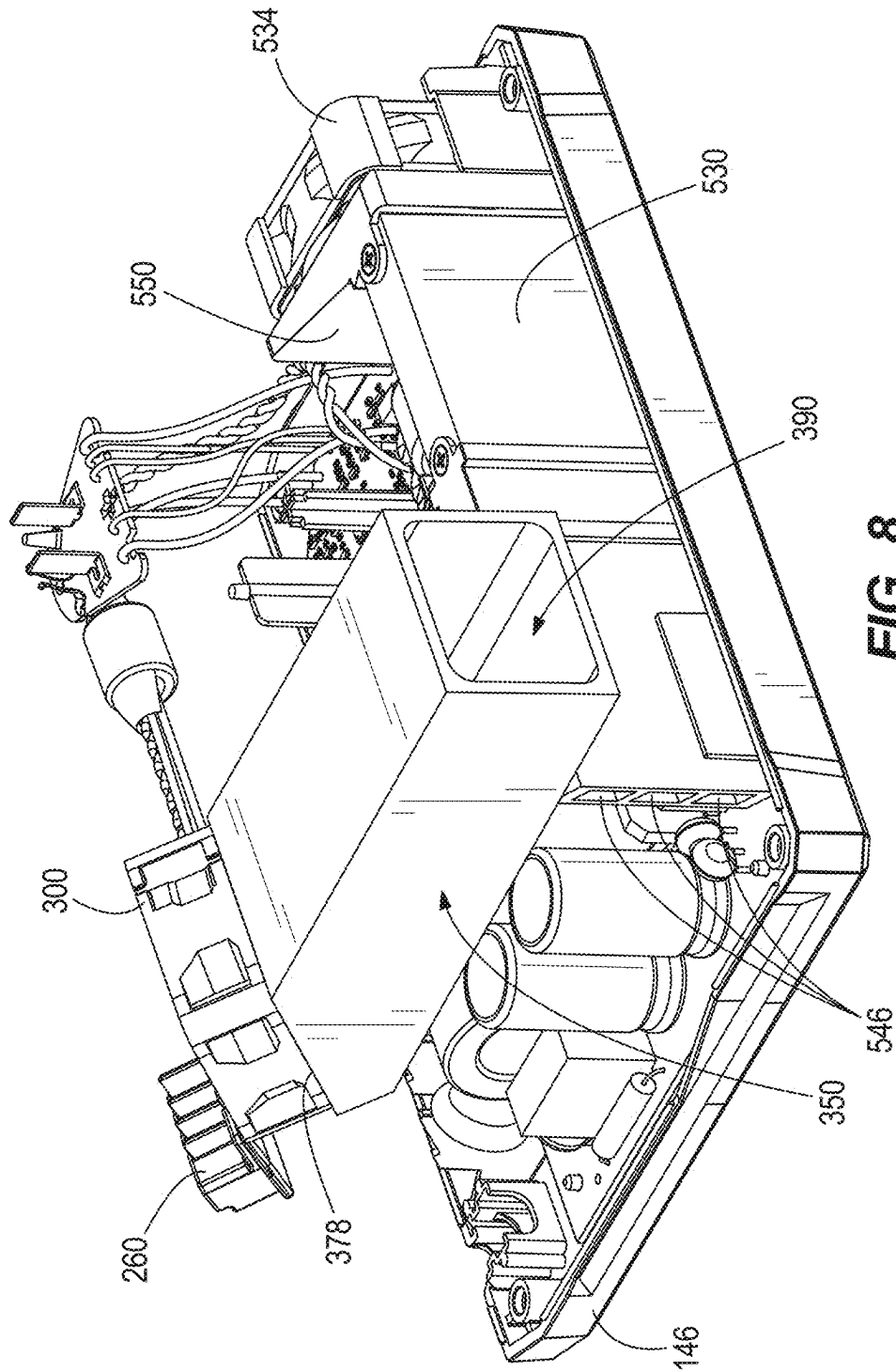


FIG. 7



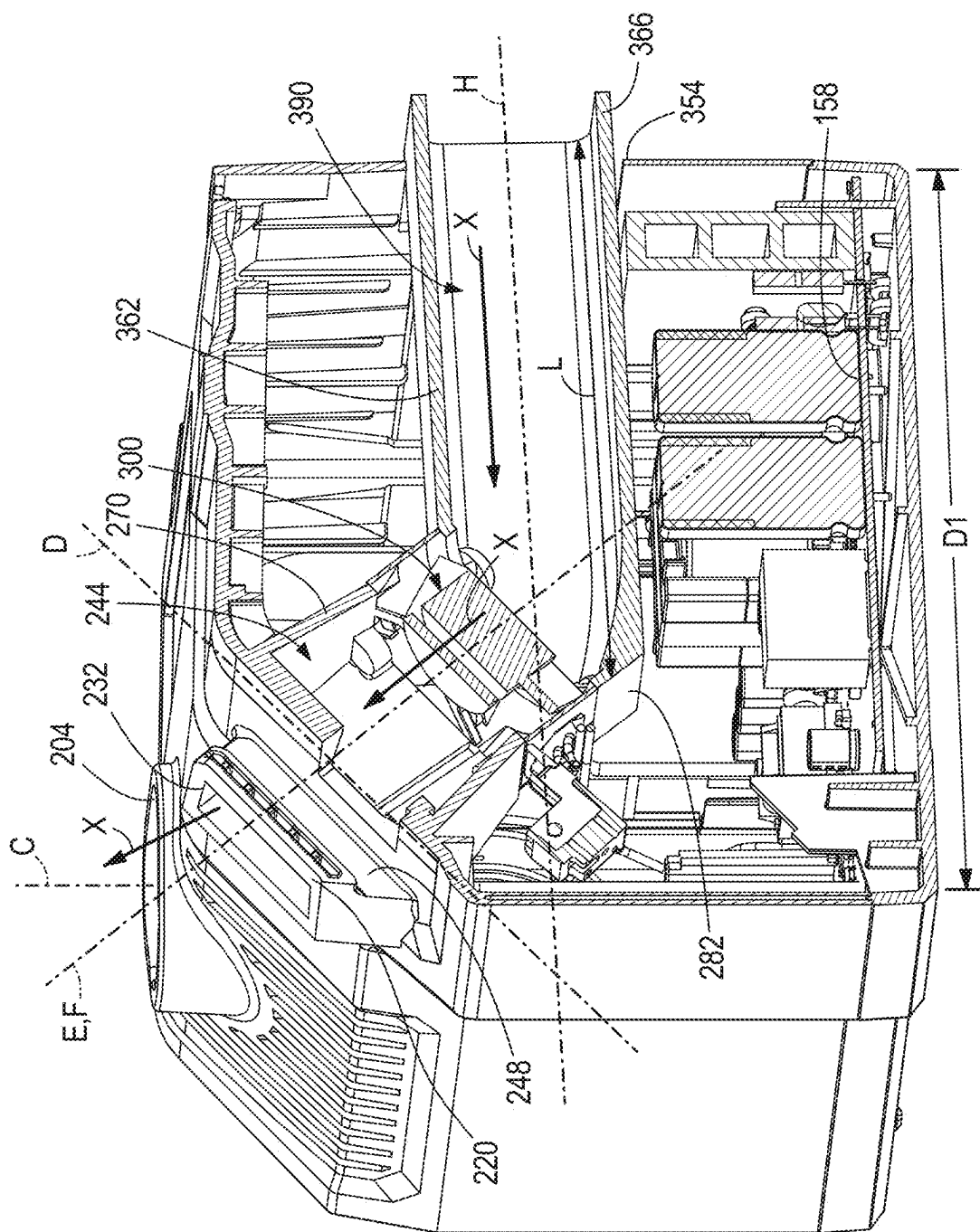
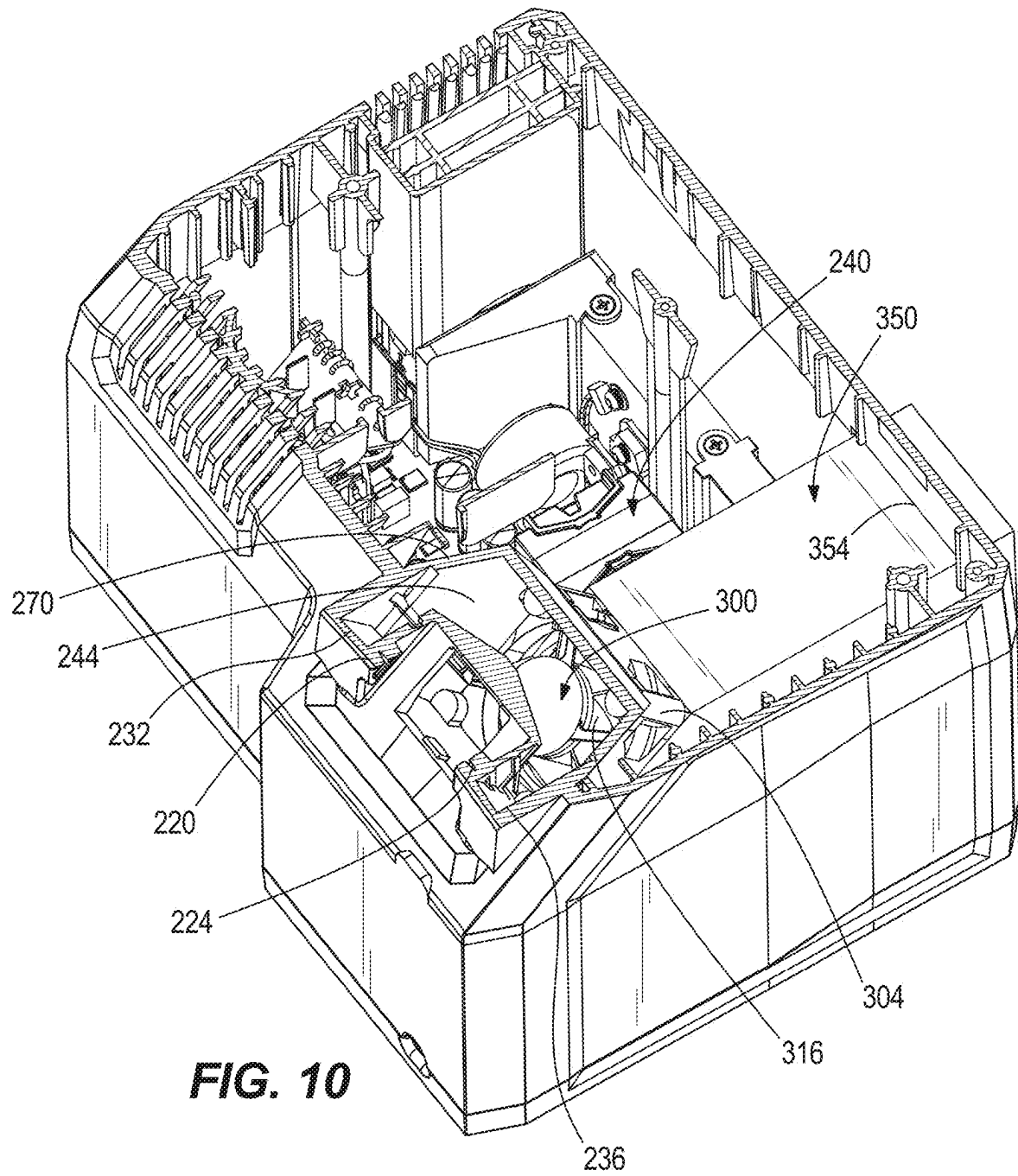
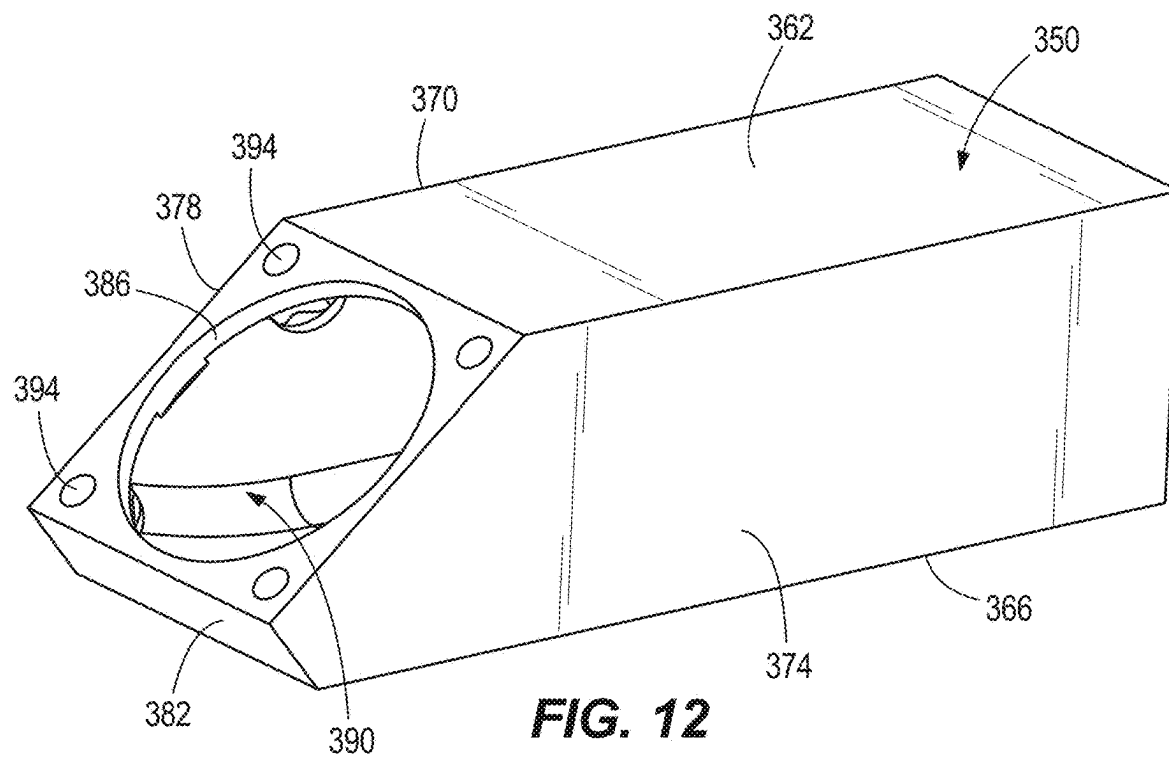
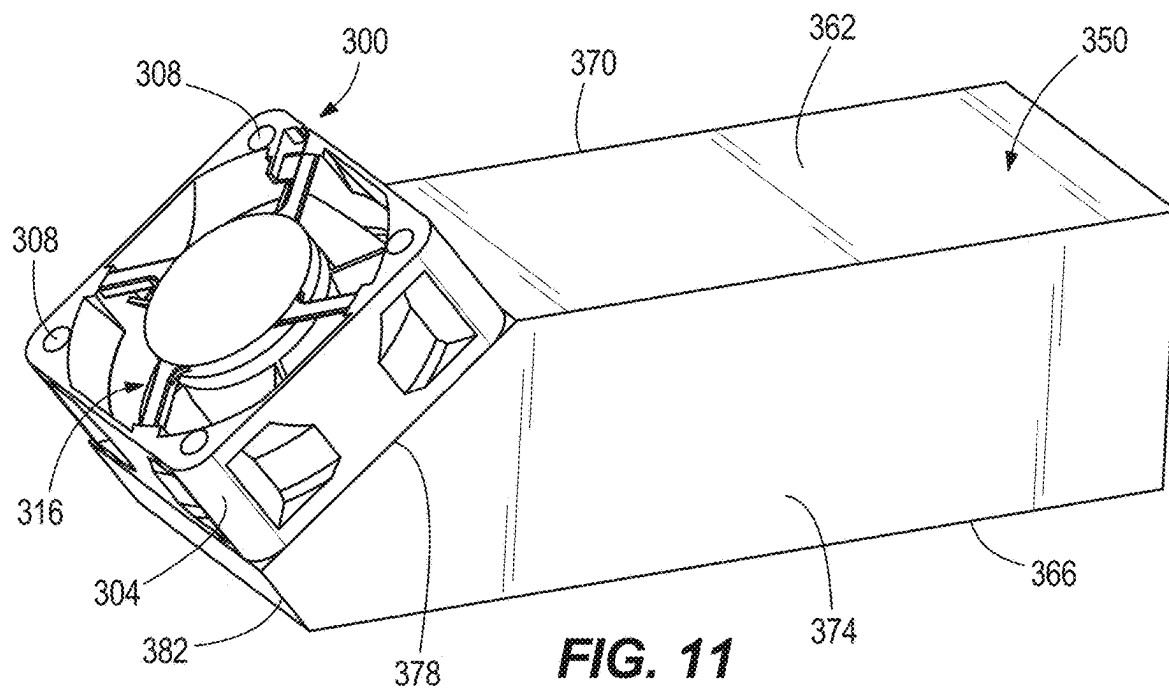
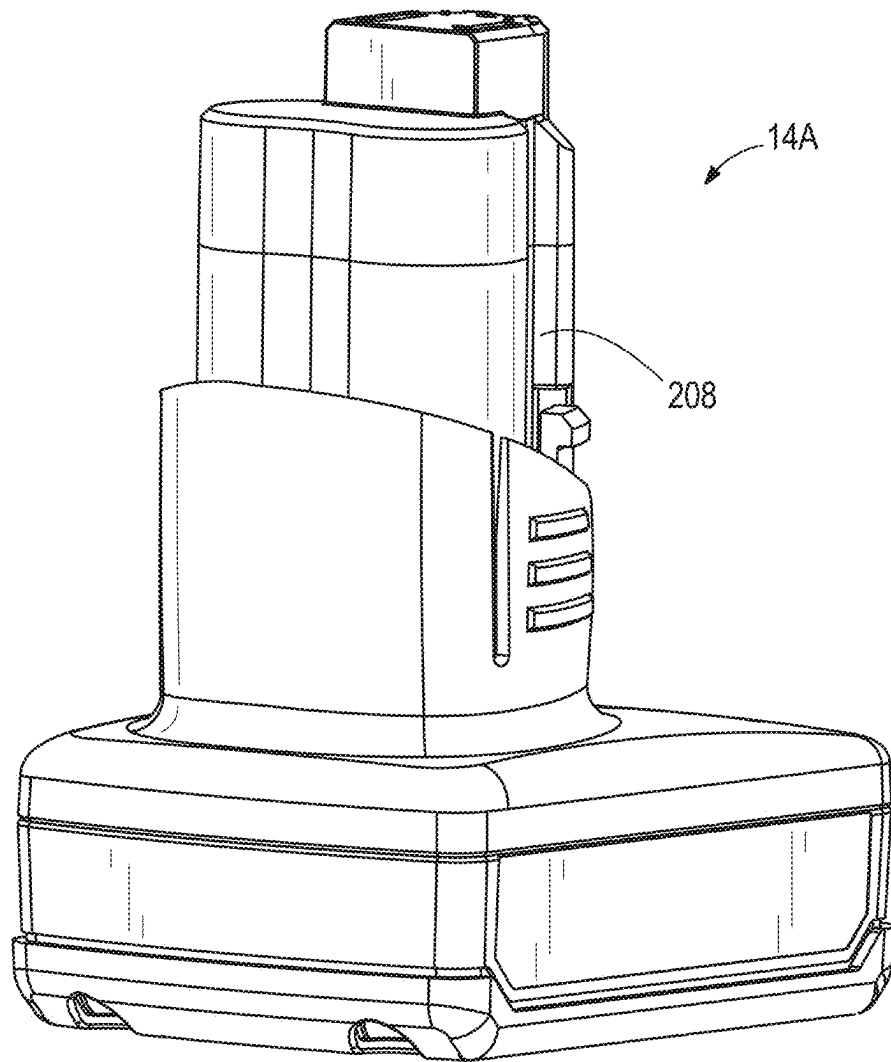
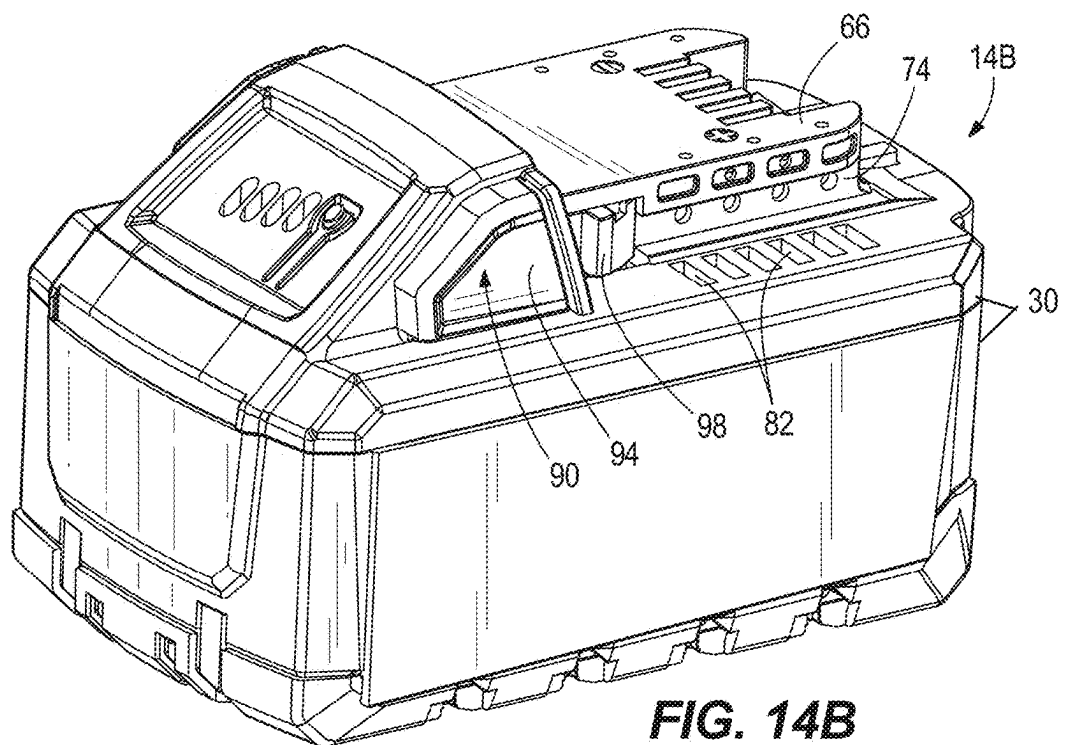
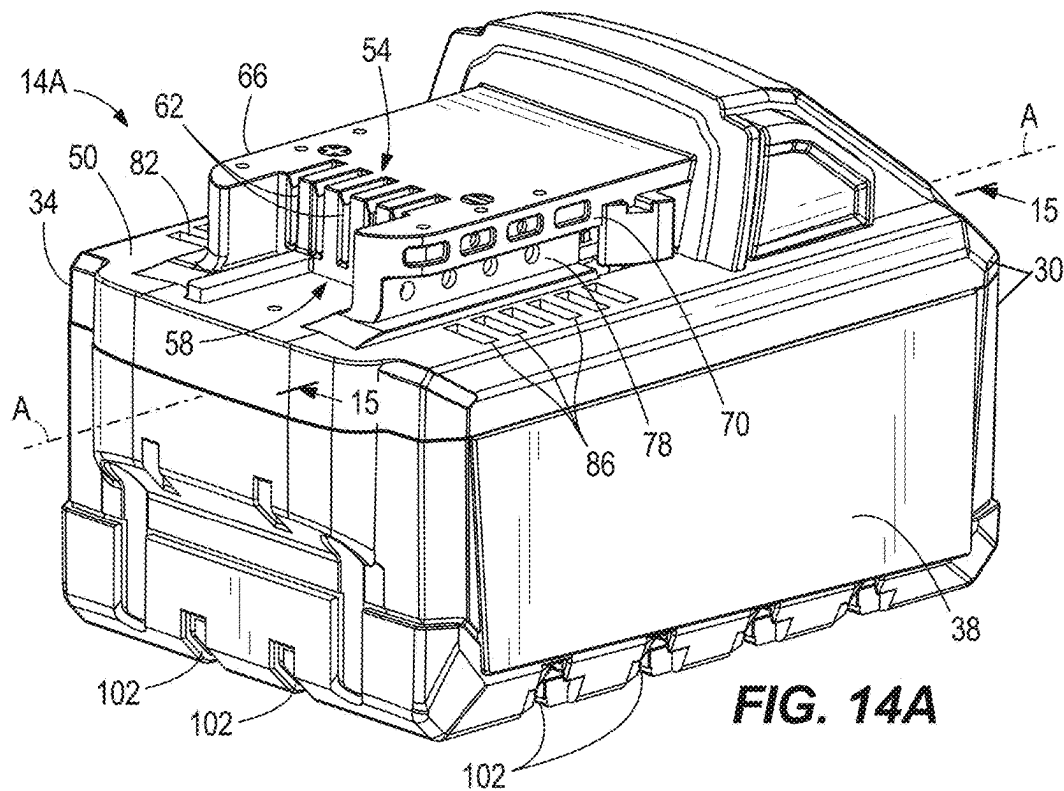


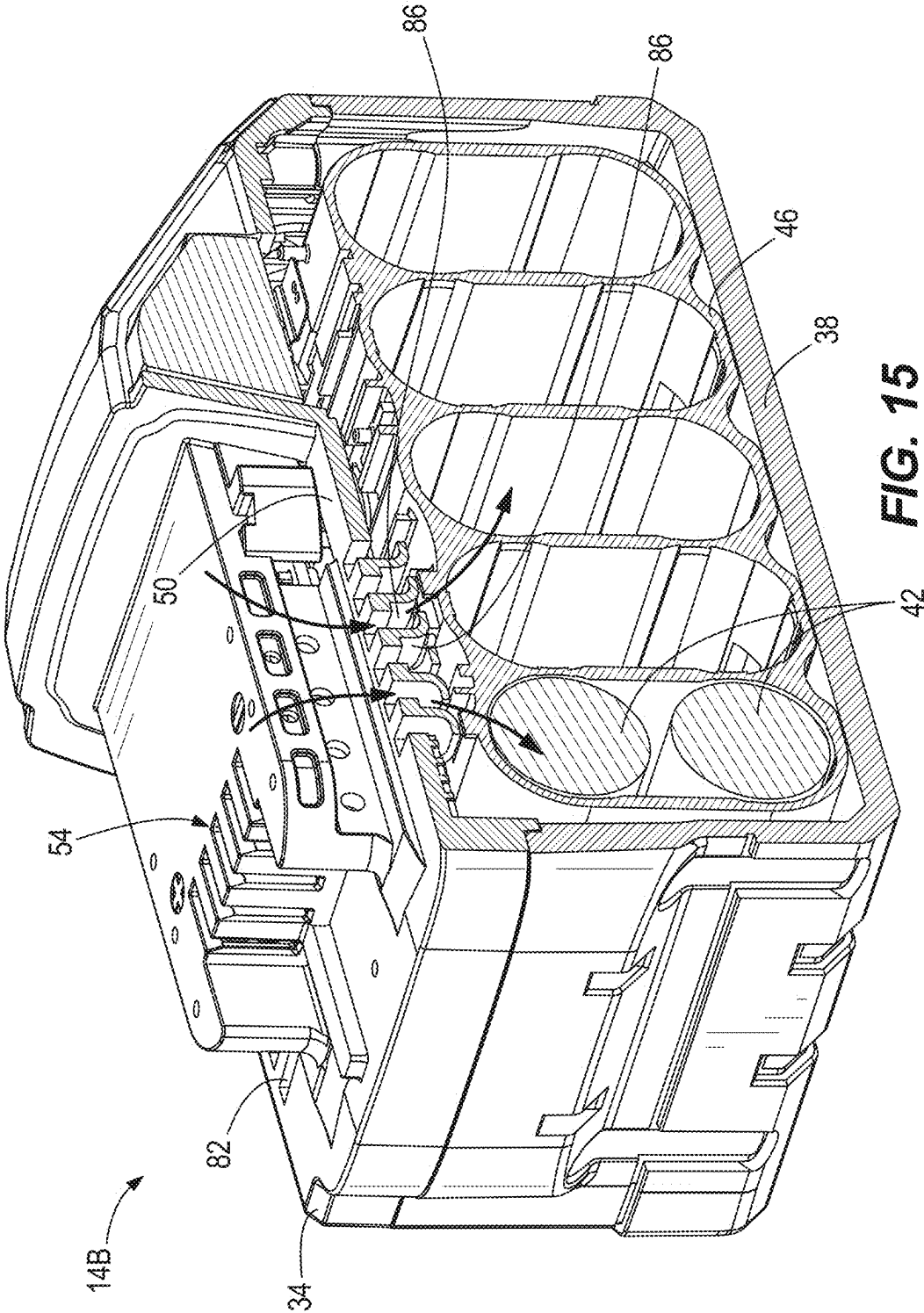
FIG. 9

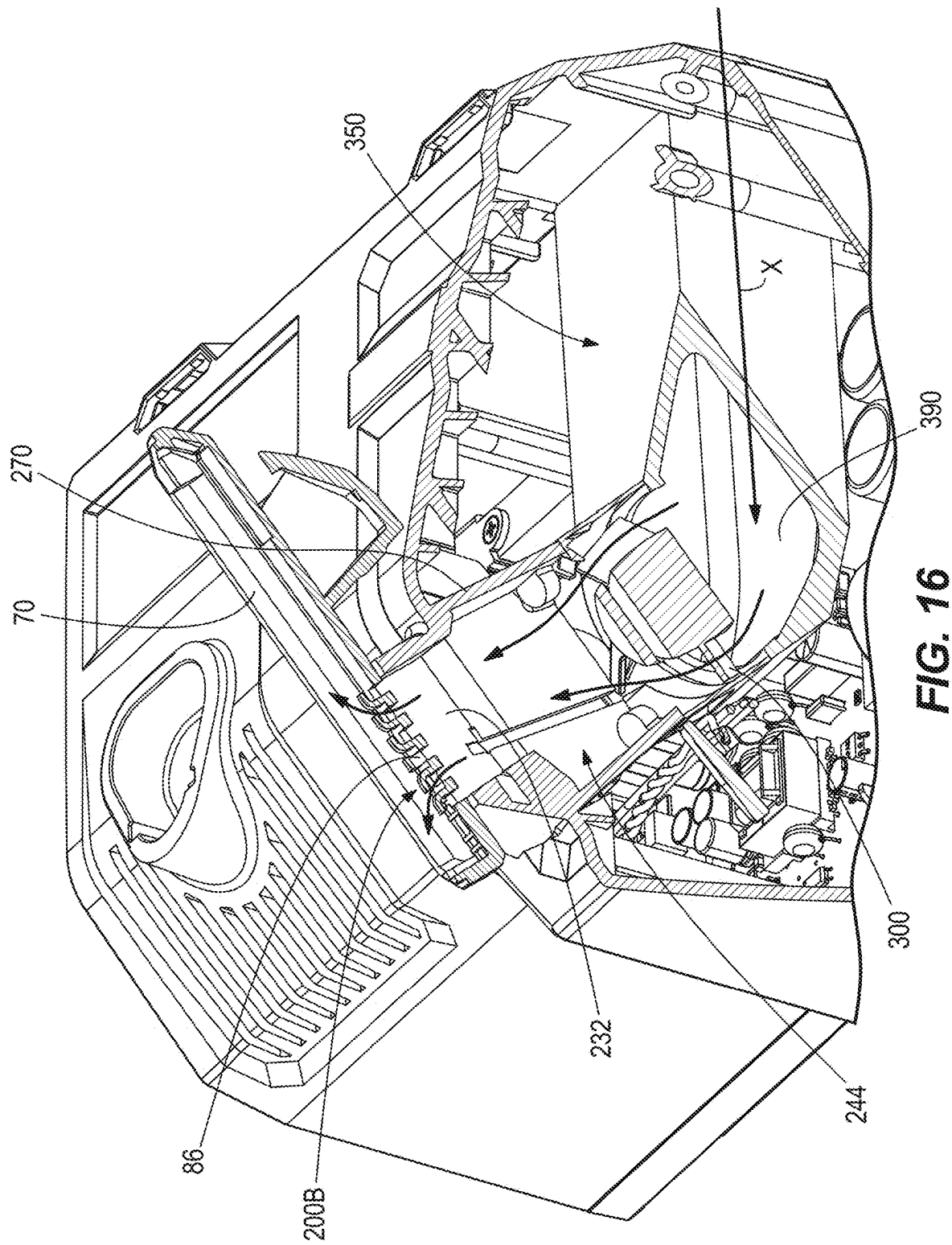




**FIG. 13**







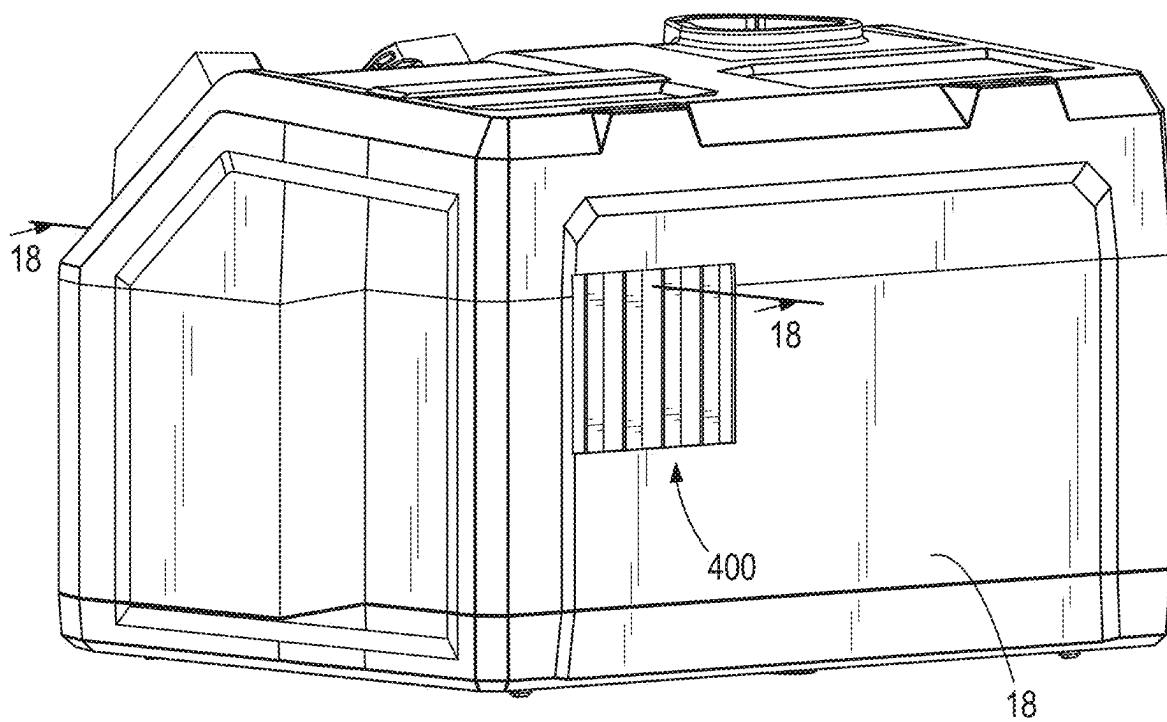


FIG. 17

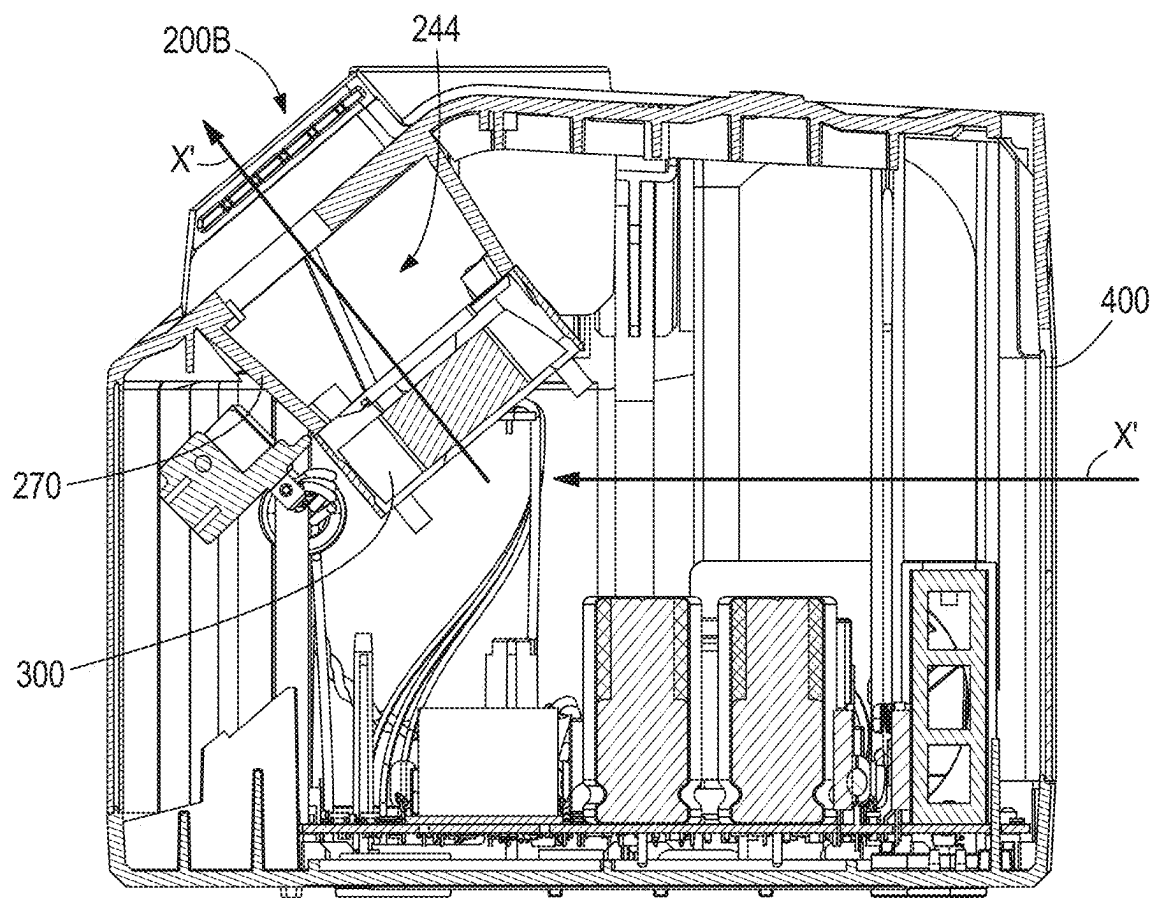


FIG. 18

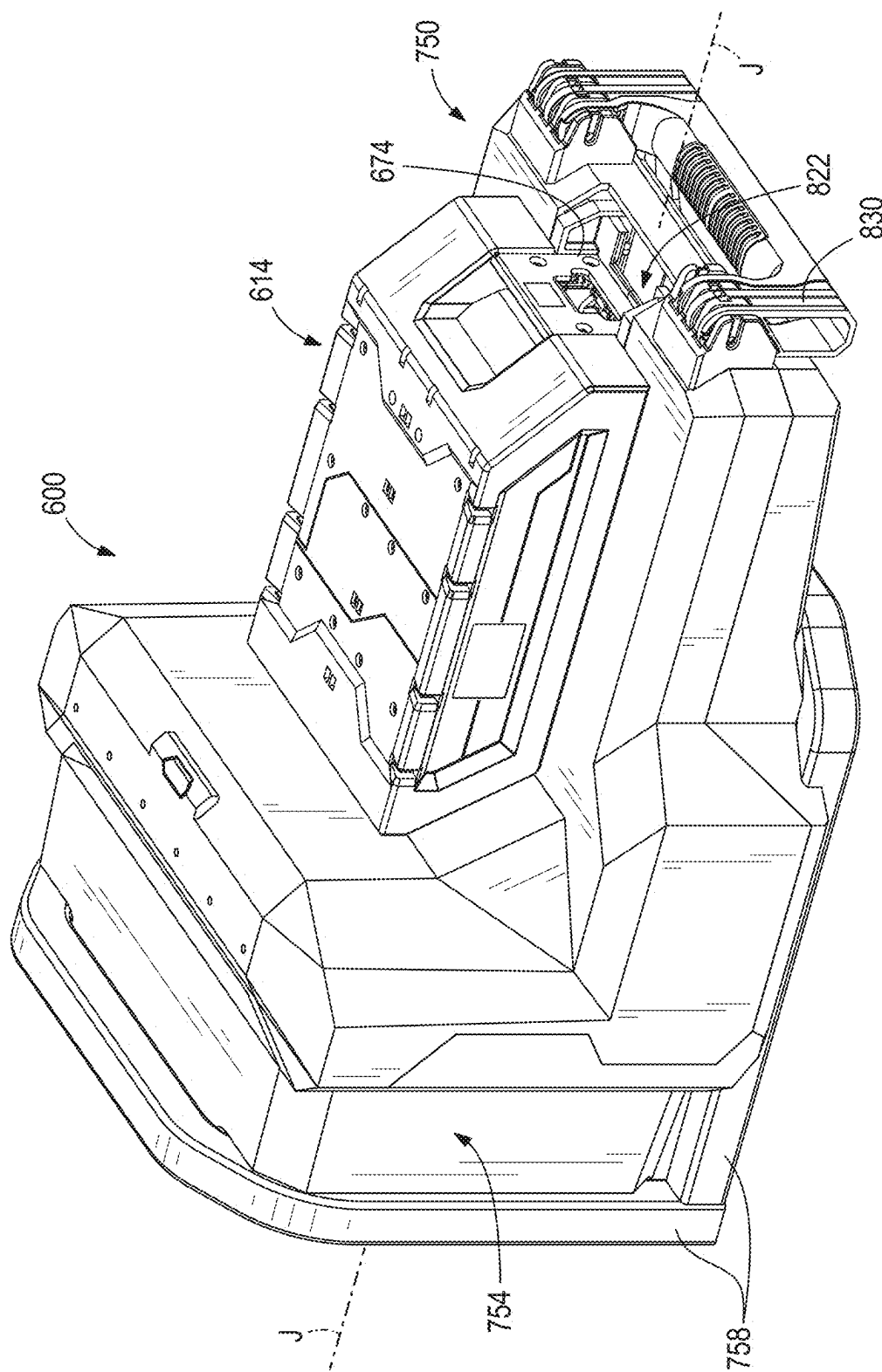
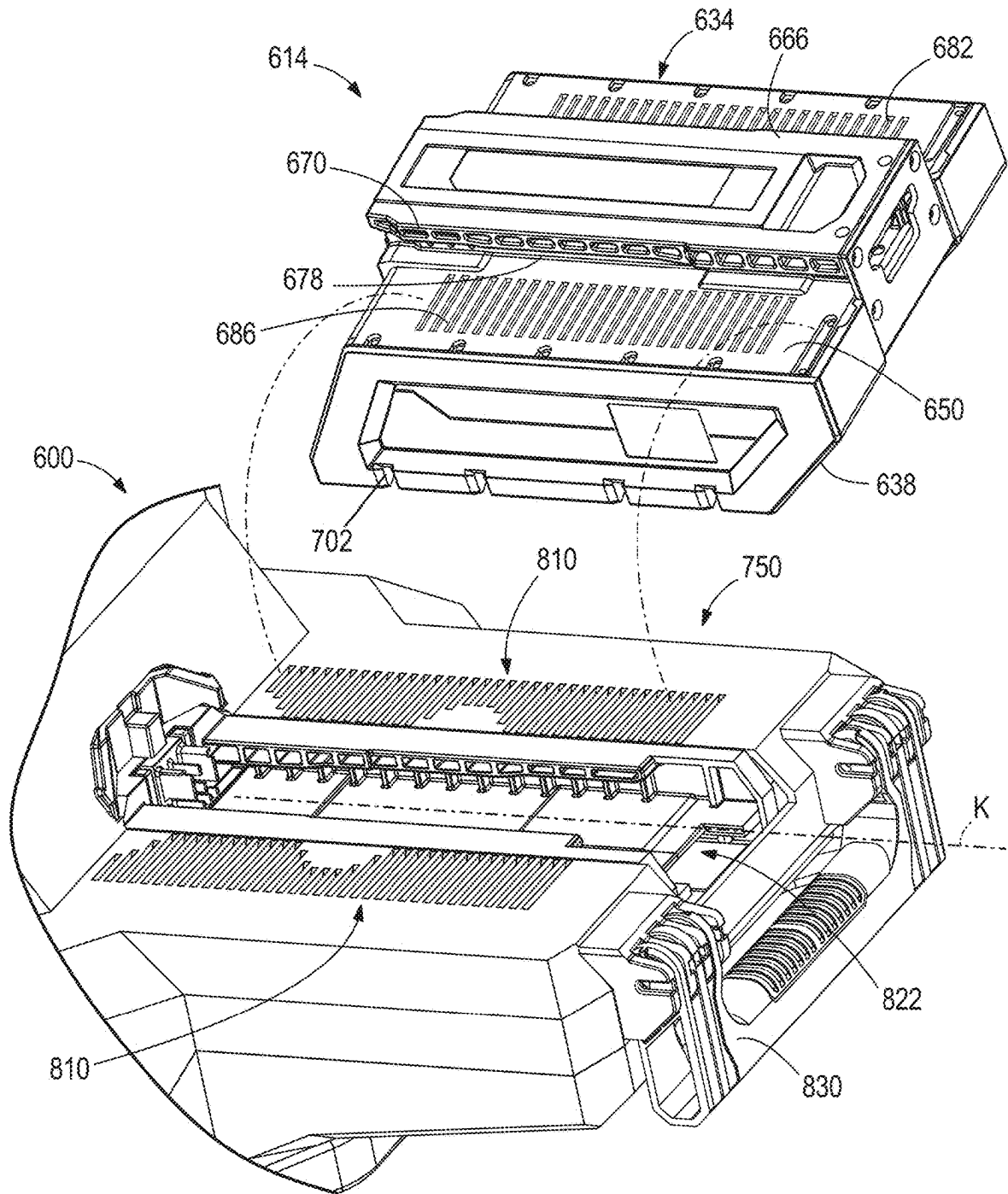
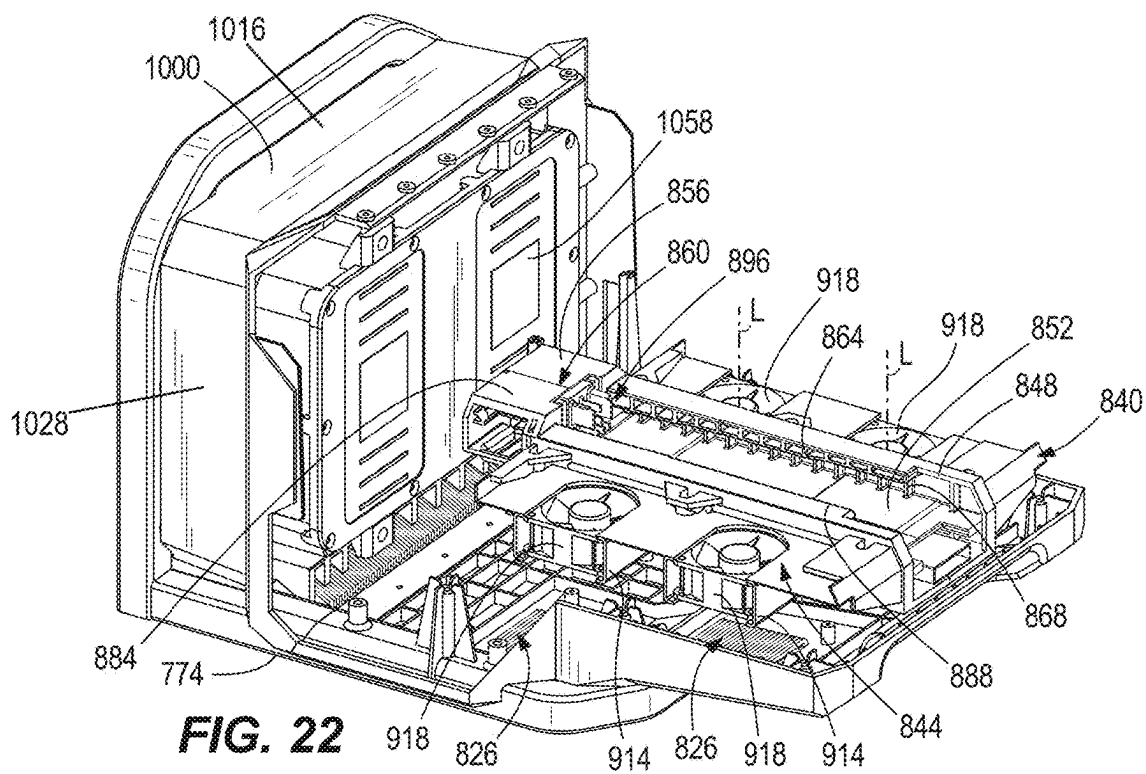
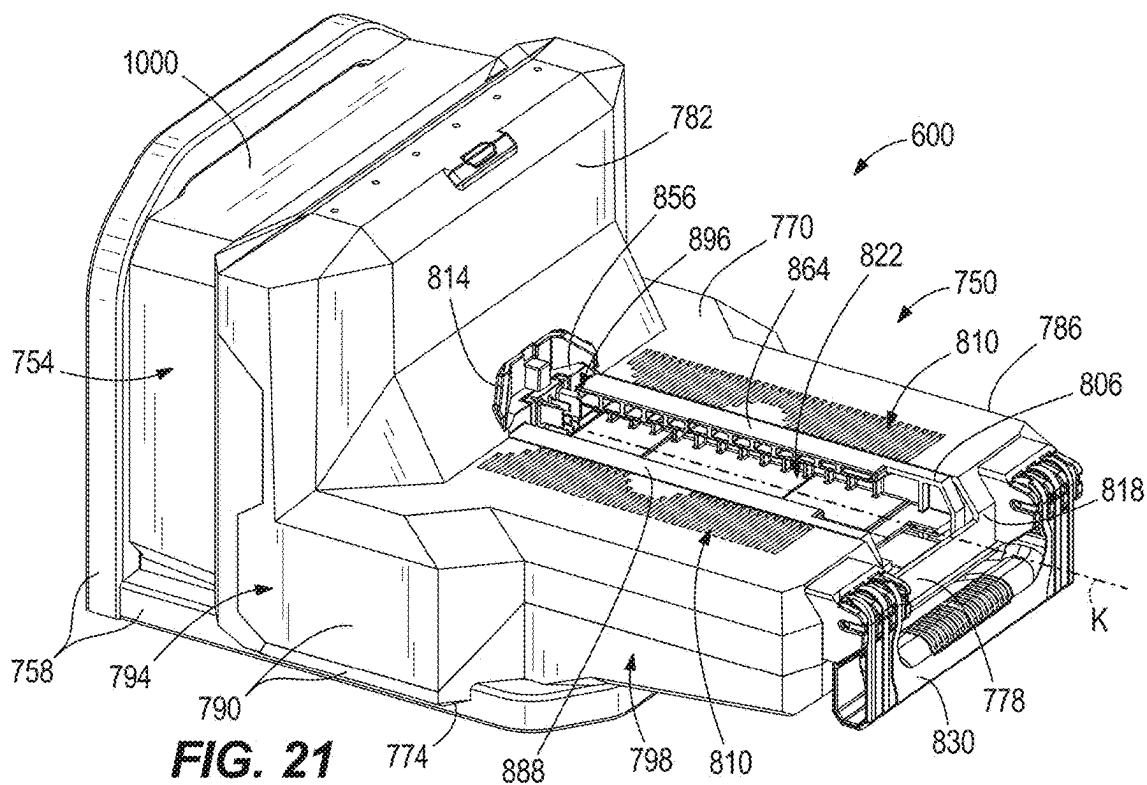


FIG. 19

**FIG. 20**



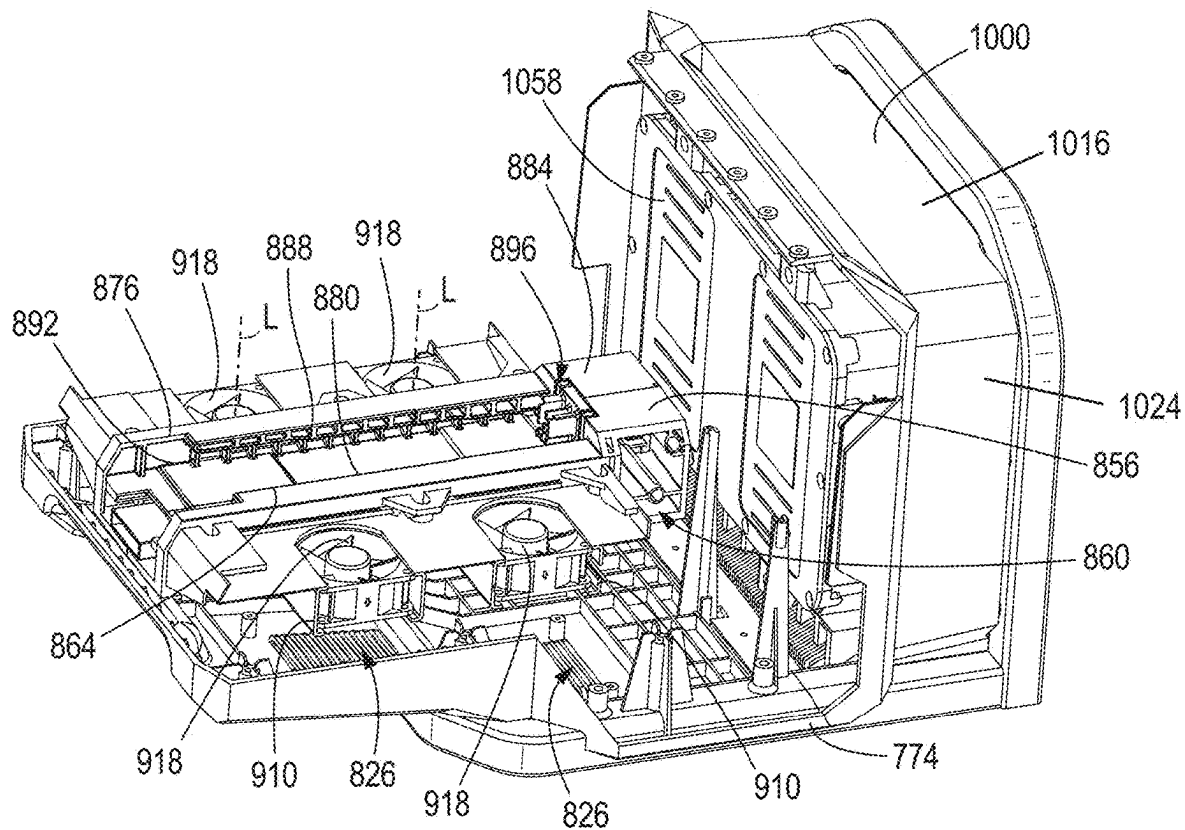


FIG. 23

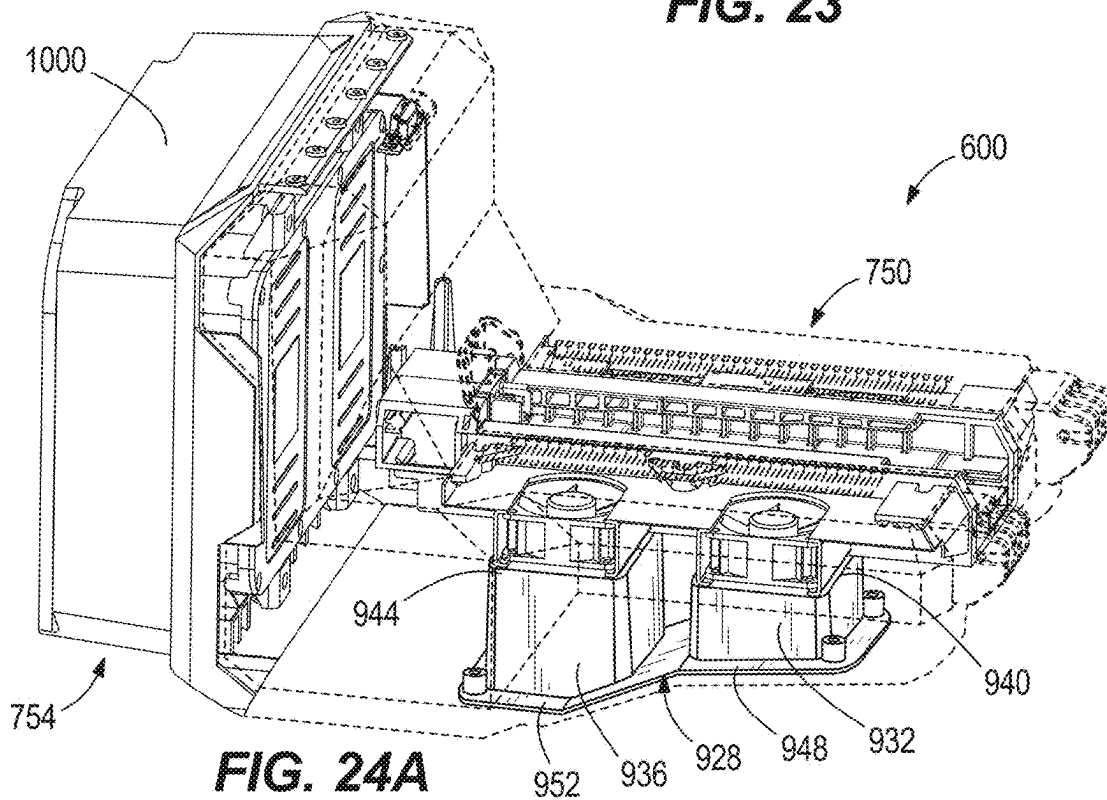
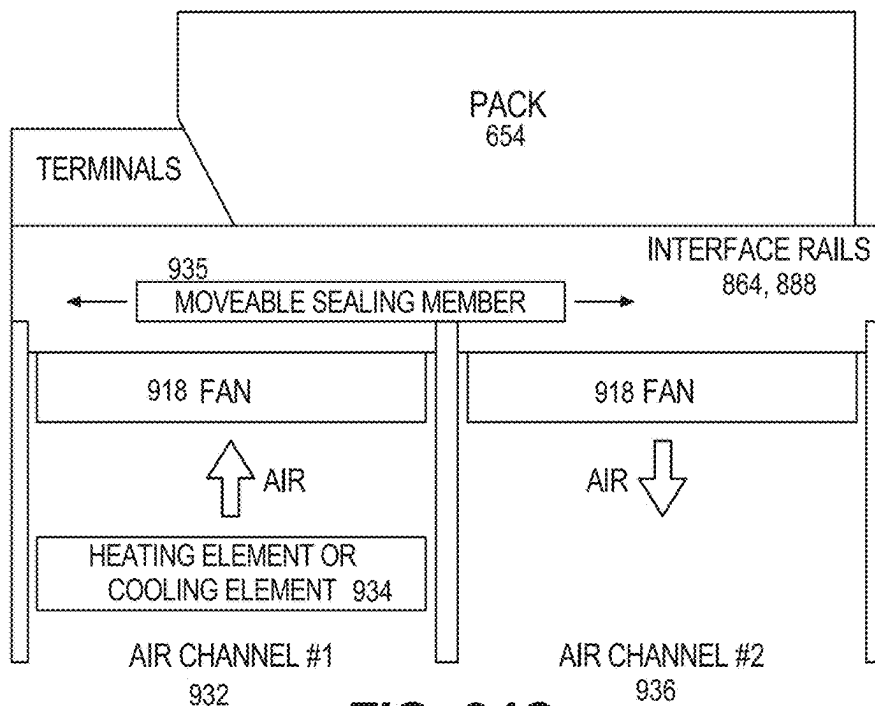
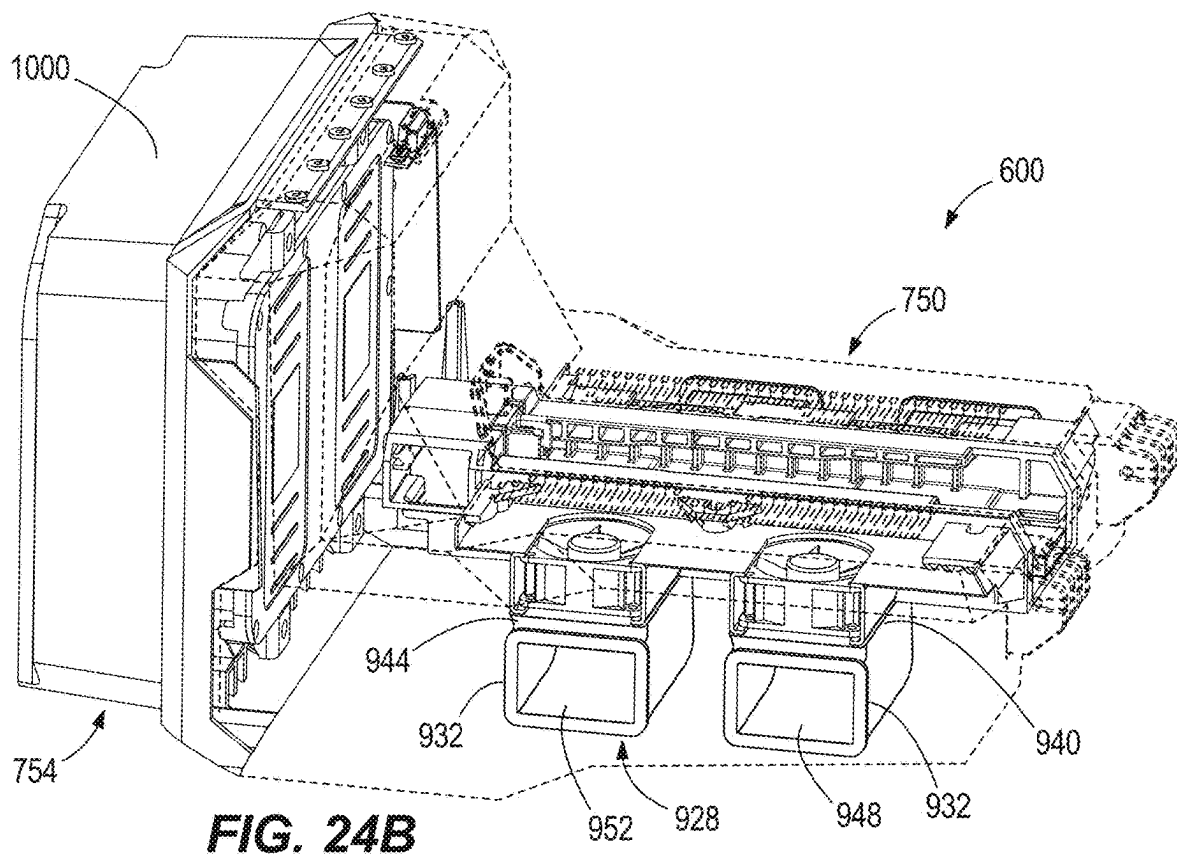
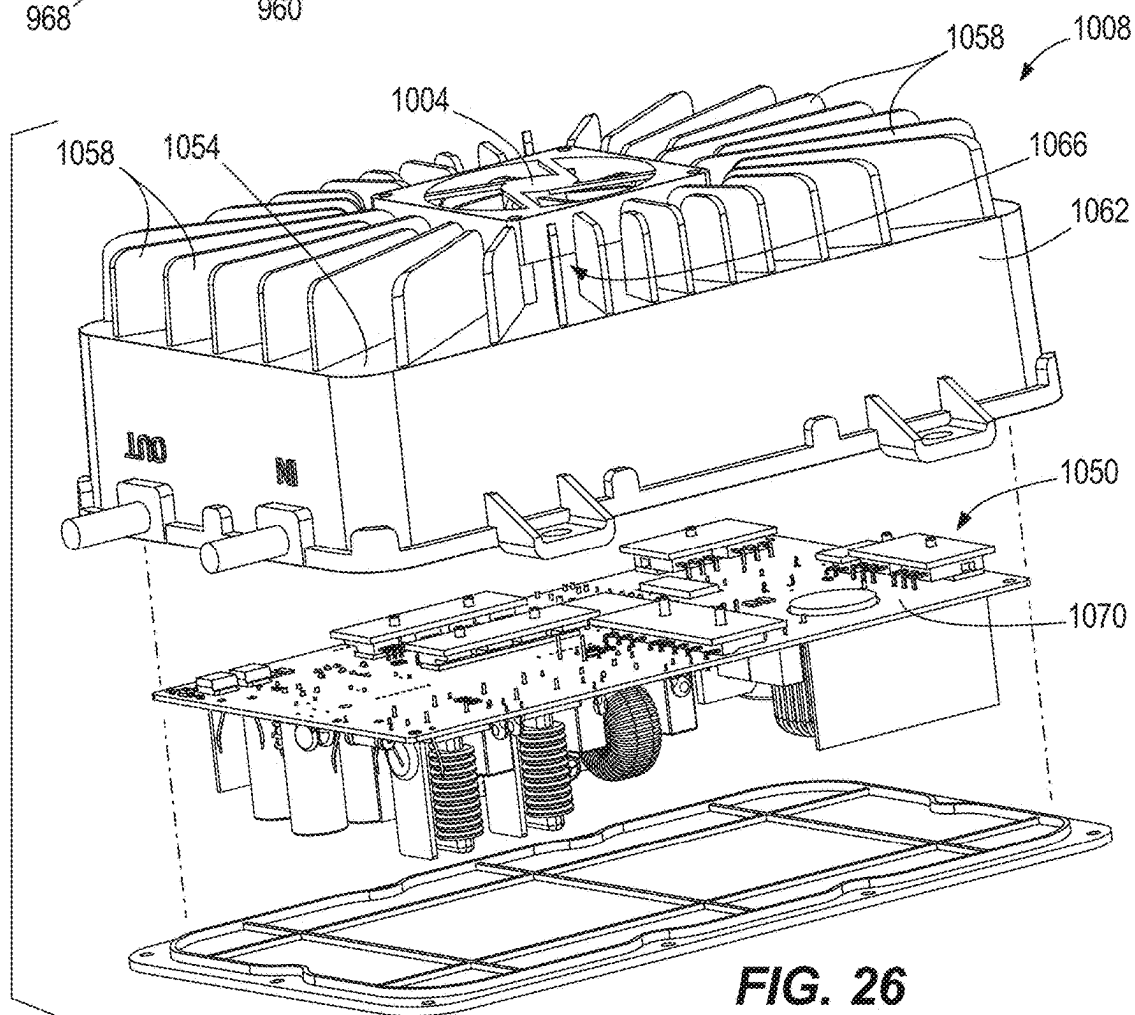
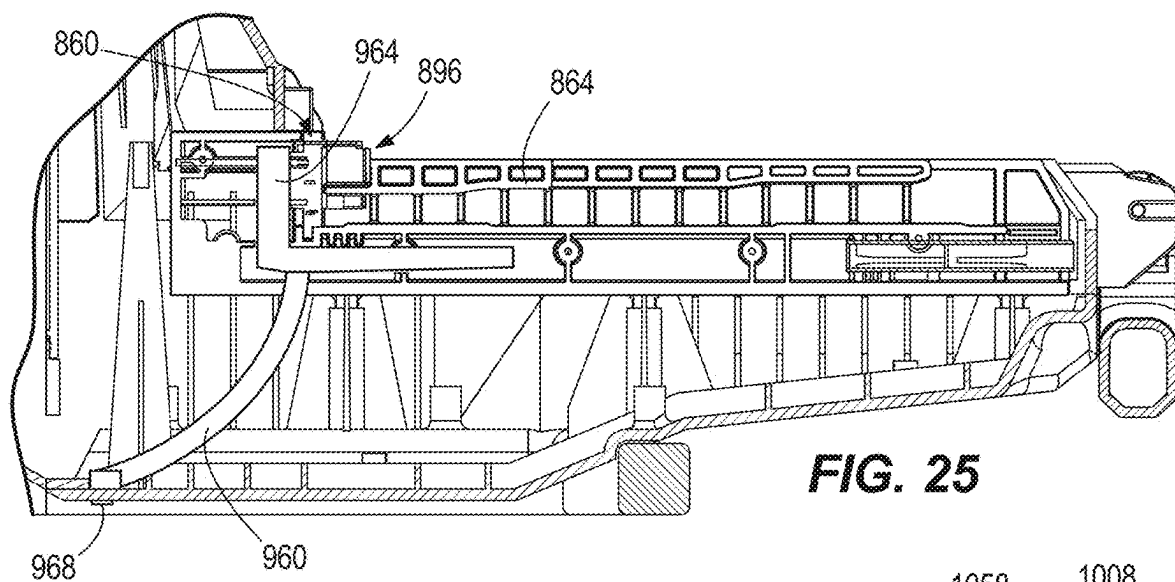


FIG. 24A





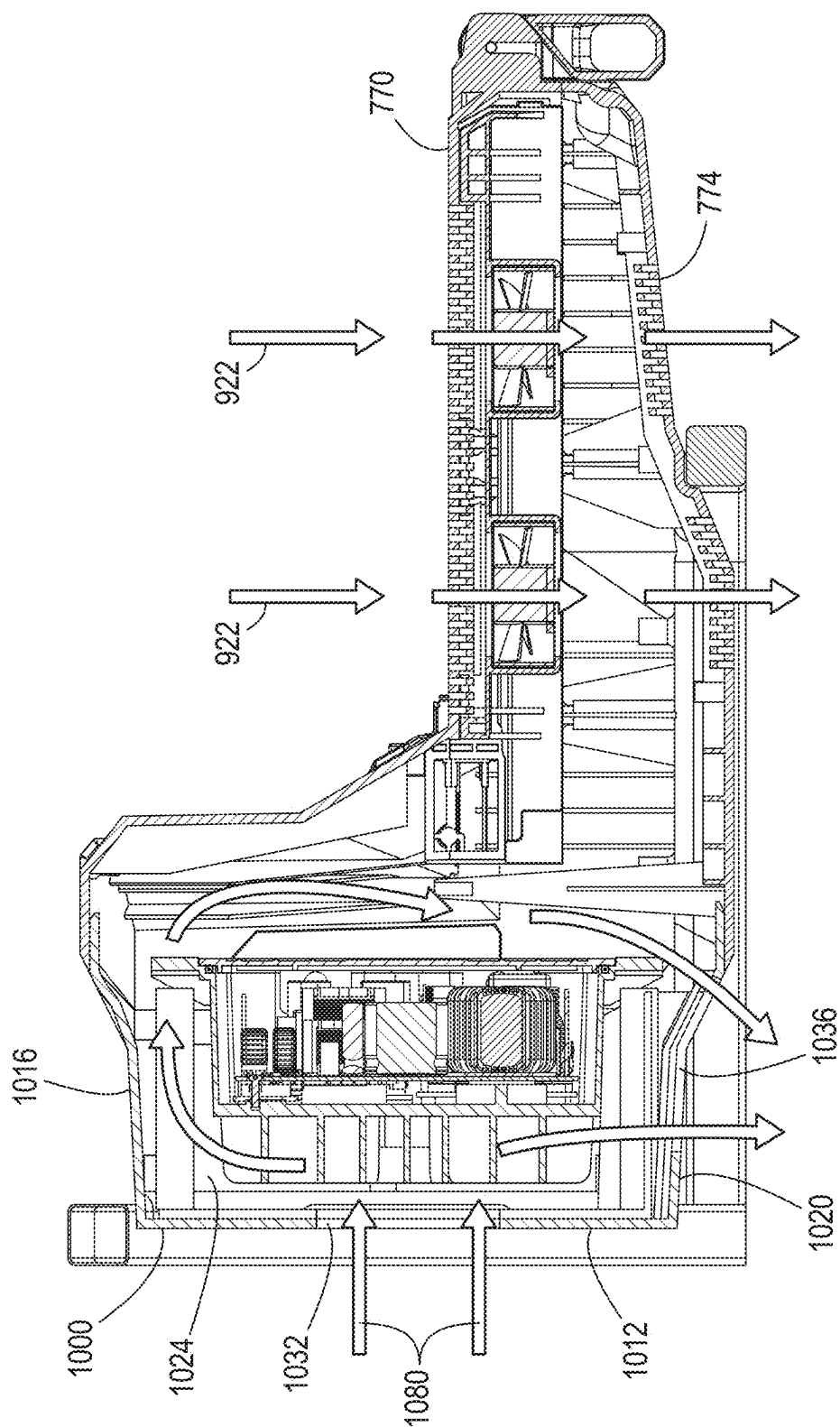
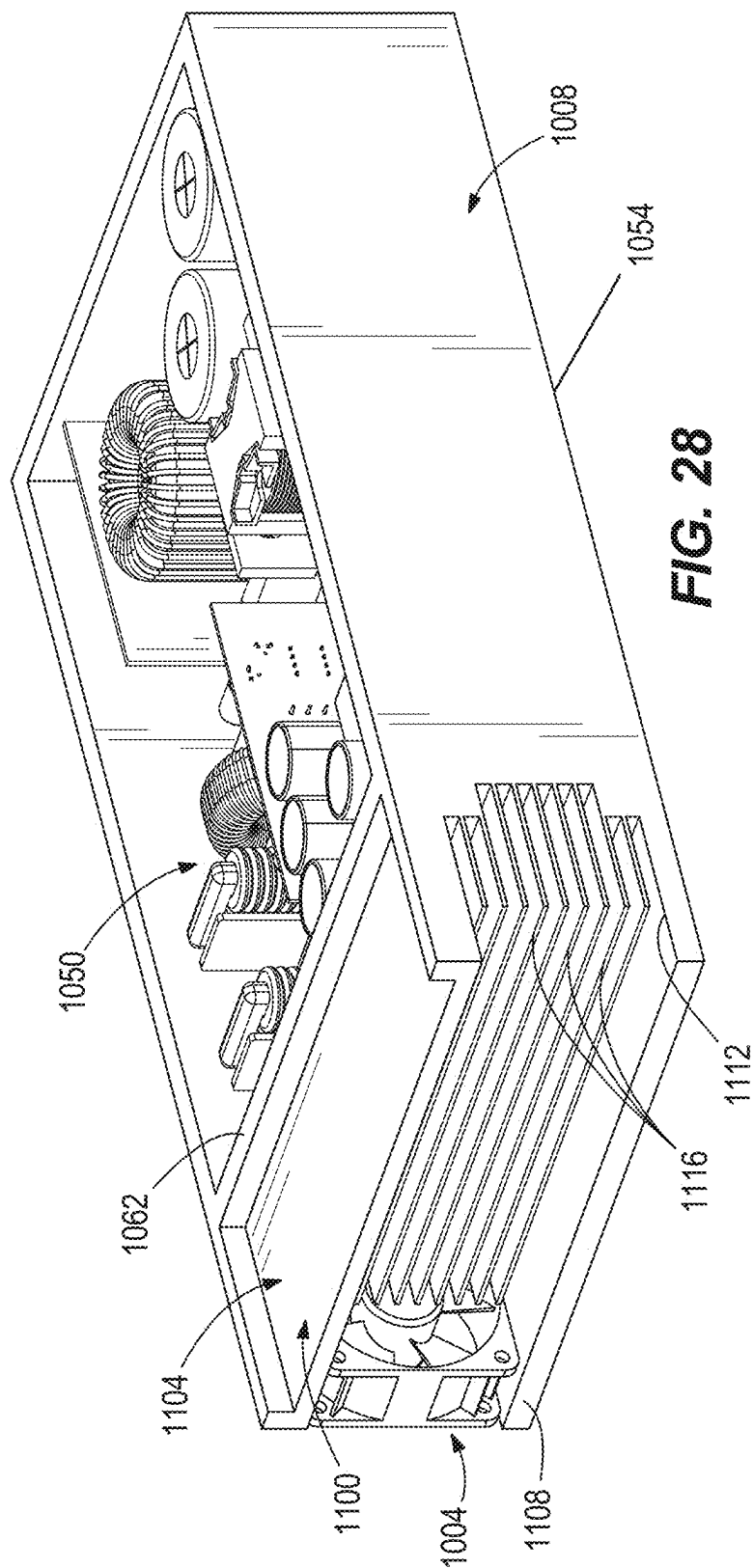


FIG. 27



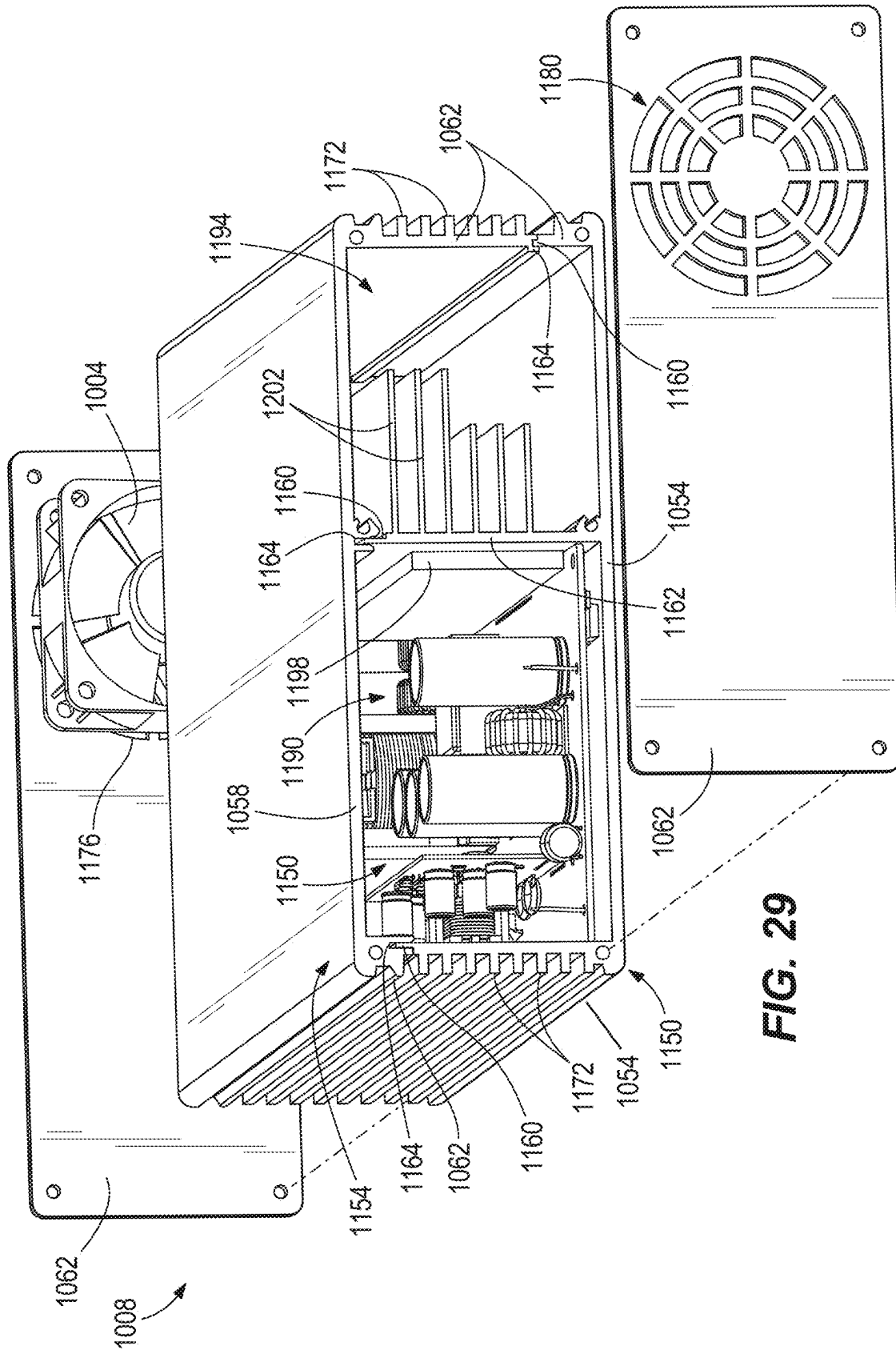


FIG. 29

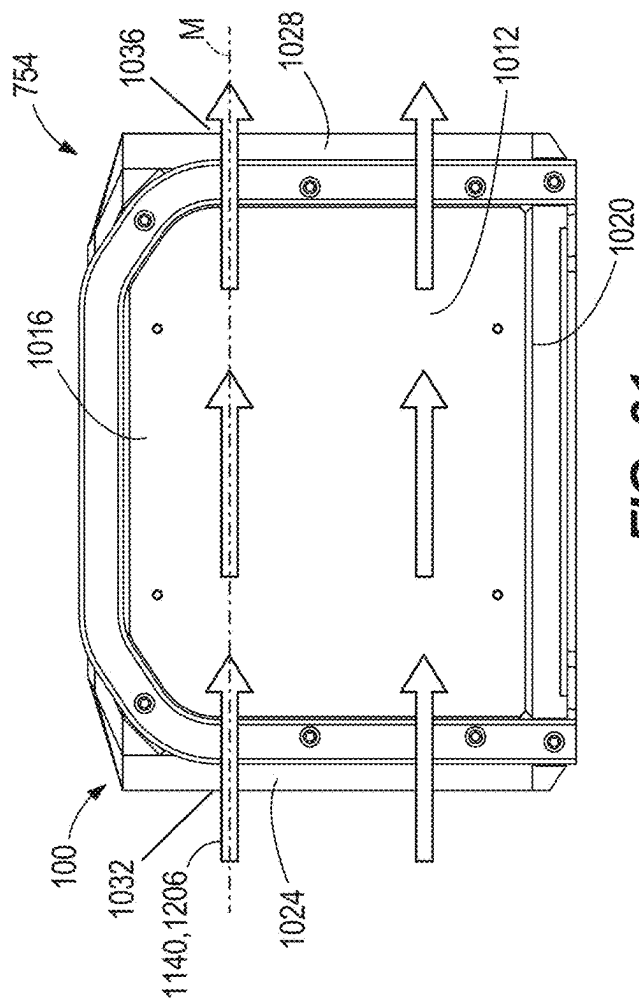


FIG. 31

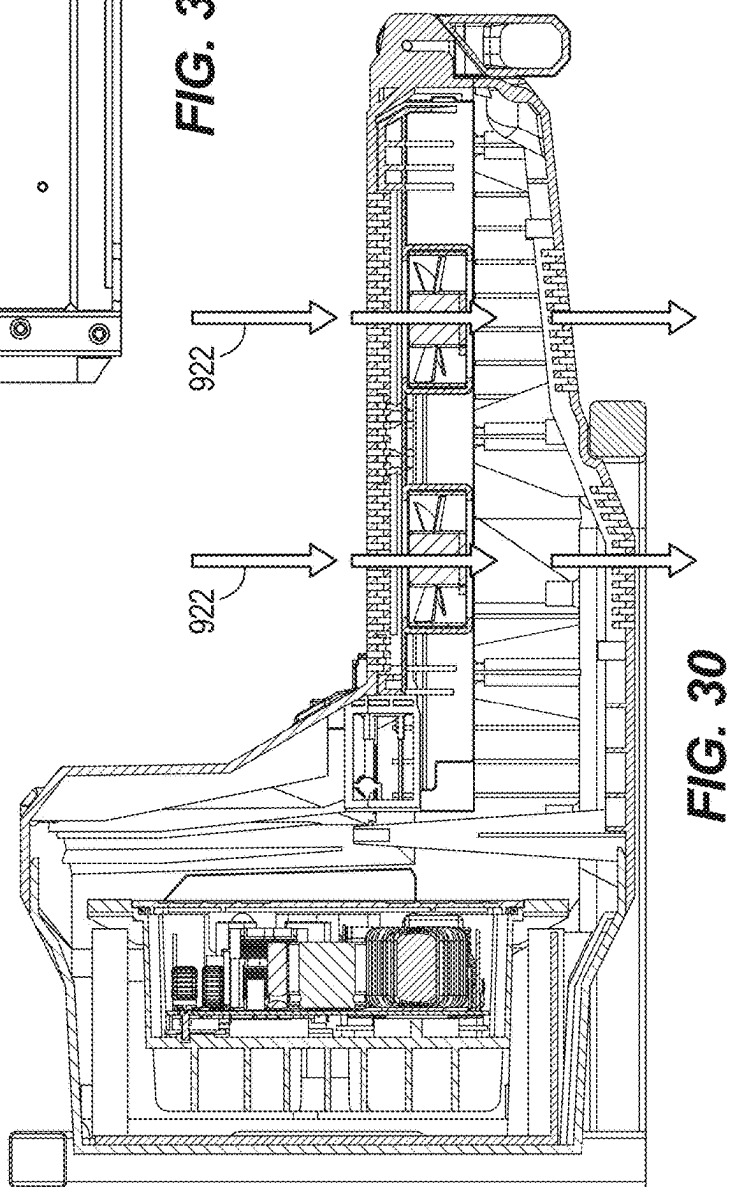


FIG. 30

BATTERY PACK AND CHARGER SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of prior-filed, co-pending U.S. patent application Ser. No. 18/308,350, filed on Apr. 27, 2023, which claims the benefit of U.S. patent application Ser. No. 17/110,456, filed on Dec. 3, 2020, which claims priority to U.S. Provisional Patent Application No. 62/942,889, filed on Dec. 3, 2019, the entire contents of each of which is incorporated by reference herein.

BACKGROUND

The present disclosure relates to battery packs and chargers for charging battery packs.

A typical battery charger includes a battery charging circuit that is connectable to a power source and to a rechargeable battery and that is operable to charge the battery.

SUMMARY

In one independent construction, a charger a housing includes a front wall, a rear wall, a top wall, a bottom wall, a first side wall, and a second side wall. An interface is positioned in the front wall and configured to engage a battery pack. The interface includes charging terminals positioned between a first rail and a second rail, a first groove positioned between the first rail and a wall of the housing, and a second groove positioned between the second rail and the wall of the housing. The interface is in communication with an interior of the housing. A fan is coupled within the housing adjacent the interface, and an air passage member includes a hollow body that has a first end and a second end spaced apart from the first end. The first end is coupled to the fan and the second end extends through one of the walls. The fan is operable to suck an air flow into the housing from outside the housing and guide the air flow through the air passage member to the interface.

In another construction, a charging system includes a charger housing having a front wall, a rear wall, a top wall, a bottom wall, a first side wall, and a second side wall. The charger housing has a first connection interface positioned in the front wall, and the interface includes charging terminals positioned between a first rail and a second rail, a first groove positioned between the first rail and a wall of the housing, and a second groove positioned between the second rail and the wall of the charger housing. Each of the first rail and the second rail include an aperture extending there-through. A fan is coupled within the housing adjacent the first connection interface. An air passage member includes a hollow body that has a first end and a second end spaced apart from the first end, the first end being coupled to the fan and the second end that extends through one of the walls. The charging system further includes a battery pack having a battery housing that has a longitudinal axis, first portion, and a second portion coupled to the first portion. The battery housing is configured to enclose a plurality of battery cells, and a second connection interface extends from a wall of the first portion and is symmetrical about the longitudinal axis. The second connection interface includes battery terminals positioned between a first rail and a second rail, a first groove positioned between the first rail and a wall of the first portion, and a second groove positioned between the second rail and the wall of the first portion. A first plurality of vent

holes extends through the wall in the first portion adjacent the first rail, and a second plurality of vent holes extends through the wall in the first portion adjacent the second rail. The second connection interface of the battery pack is coupleable to the first connection interface such that aperture in the first rail of the first connection interface is in communication with the first plurality of vent holes and the aperture in the second rail of the first connection interface is in communication with the second plurality of vent holes. The fan is operable to suck an air flow into the housing of the charger from outside the housing and guide the air flow through the air passage member to the apertures in each of the first and second rails and the first and second plurality of vent holes such that the air flow enters the battery pack housing.

In another construction, a charger for charging a battery pack having a housing enclosing a plurality of battery cells and one or more vent holes includes a housing including a battery pack receiving portion and a separate charger electronics portion. The charger electronics portion includes a charger electronics enclosure. The charger further includes a battery pack interface supported on the battery pack receiving portion, and the battery pack interface includes charging terminals. A first plurality of vent holes is positioned in a first outer wall of the battery pack receiving portion adjacent the battery pack interface. A first cooling air passage extends through the battery pack receiving portion from the first plurality of vent holes to an opening in a second outer wall of the battery pack receiving portion, and a first fan is positioned in the battery pack receiving portion adjacent the first plurality of vent holes and is operable to move air from outside of the battery pack receiving portion into the first cooling air passage via the first plurality of vent holes. The charger further includes a printed circuit board including electrical components in electrical communication with the charging terminals. The printed circuit board is enclosed within charger electronics enclosure. A second plurality of vent holes is positioned in a third outer wall of the charger electronics portion. A second cooling air passage extends through the charger electronics portion from the second plurality of vent holes to an opening in a fourth outer wall of the charger electronics portion, and the second cooling air passage is in fluid communication with the charger electronics enclosure for removing heat generated by the electrical components and sealed from the first cooling air passage. A second fan is positioned in the charger electronics portion outside the charger electronics enclosure adjacent the second plurality of vent holes and operable to move air from outside charger electronics portion into the second cooling air passage via the second plurality of vent holes.

Other independent aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a battery charger according to a first construction.

FIG. 2 is another perspective view the charger of FIG. 1.

FIG. 3 is a top view of the charger of FIG. 1.

FIG. 4 is a bottom perspective of the charger of FIG. 1.

FIG. 5 is a rear perspective of the charger of FIG. 1.

FIG. 6 is a bottom perspective view of a portion of a housing of the charger of FIG. 1.

FIG. 7 is a top perspective view of the charger of FIG. 1, with portions of the housing removed and showing an air passage.

3

FIG. 8 is another top perspective view of the charger of FIG. 1, with portions of the housing removed and showing an air passage and a fan.

FIG. 9 is a cross-sectional view of the charger of FIG. 1 along the lines 9-9 of FIG. 3.

FIG. 10 is another cross-sectional view of the charger of FIG. 1 along the lines 10-10 of FIG. 5.

FIG. 11 is a perspective view of the air passage and the fan of FIGS. 7 and 8.

FIG. 12 is a perspective view of the air passage of FIGS. 7 and 8.

FIG. 13 is a perspective view of a battery pack of a first type.

FIG. 14A is a rear perspective view of a battery pack of a second type.

FIG. 14B is a front perspective view of a battery pack of FIG. 14A.

FIG. 15 is a cross-sectional view of the battery pack of FIGS. 14A-14B along the line 15-15 of FIG. 14A.

FIG. 16 is a cross-sectional view of the battery pack of FIG. 14A coupled to the charger of FIG. 1 along the line 16-16 of FIG. 1.

FIG. 17 is a perspective view of a battery charger according to another construction.

FIG. 18 is a cross-sectional view of the battery charger along the line 18-18 of FIG. 17.

FIG. 19 is a perspective view of a battery charger according to another construction coupled to a complementary battery pack.

FIG. 20 is an exploded view of the battery charger and the battery pack of FIG. 19. FIG. 21 is a perspective view of the battery charger of FIG. 19.

FIG. 21 is a perspective view of the battery charger of FIG. 19.

FIG. 22 is a perspective view of the battery charger of FIG. 19 with a portion of the housing removed and showing a fan orientation.

FIG. 23 is another perspective view of the battery charger of FIG. 19 with a portion of the housing removed and showing a fan orientation.

FIG. 24A is a perspective view of the battery charger with a portion of the housing removed and showing another fan orientation.

FIG. 24B is a perspective view of the battery charger with a portion of the housing removed and showing another fan orientation.

FIG. 24C is a schematic view of a battery charger having an air passage member with dual air passages.

FIG. 25 is a cross-sectional view of the battery charger of FIG. 19 along an insertion axis K, shown in FIG. 21.

FIG. 26 is an exploded view of a charger electronics portion of the housing of the battery charger of FIG. 19.

FIG. 27 is a side view of the battery charger of FIG. 19 illustrating cooling air passages.

FIG. 28 is a perspective view of a charger electronics portion of the housing of the battery charger of FIG. 19 according to another construction.

FIG. 29 is an exploded view of a charger electronics portion of the housing of the battery charger of FIG. 19 according to another construction.

FIG. 30 is a side view of the battery charger of FIG. 19 illustrating a first set of cooling air passages.

FIG. 31 is a rear view of the battery charger of FIG. 19 illustrating a second set of cooling air passages through the charger electronics housing of either FIGS. 28 and 29.

DETAILED DESCRIPTION

Before any independent constructions of the disclosure are explained in detail, it is to be understood that the

4

disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other independent constructions and of being practiced or of being carried out in various ways.

Use of “including” and “comprising” and variations thereof as used herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Use of “consisting of” and variations thereof as used herein is meant to encompass only the items listed thereafter and equivalents thereof.

FIGS. 1-5 illustrate a charging system according to a first construction and including a battery charger 10 operable to charge a battery pack 14A, 14B (FIGS. 13-15) coupled to the charger 10. In the illustrated construction, the battery charger 10 is operable to charge a first battery pack 14A (FIG. 13) of a first type and a second battery pack 14B (FIGS. 14A-15) of a second type. The illustrated battery charger 10 may be operable to charge a high-output battery pack (e.g., having a current capacity of 12 amp-hours (Ah) or more), which requires about three times the power of typical chargers, in about 60 minutes.

With reference to FIGS. 13-15, the battery pack type may be defined by nominal voltage, current capacity, connection configuration (e.g., “tower” vs. “slide-on”), etc., of the battery pack 14A, 14B. For example, the first battery pack 14A may include a high-power battery pack with a nominal voltage of about 12 volts (V) and having a tower-style configuration, and the second battery pack 14B may include a high-power battery pack with a nominal voltage of 18V and a slide-on configuration. In other constructions (not shown), the battery packs 14A, 14B may be the same type of battery pack.

As shown in FIGS. 14A-15, the battery pack 14B may include a battery pack housing 30 including a longitudinal axis A, a first (e.g., top) portion 34, and a second (e.g., bottom) portion 38 that is coupled to the first portion 34. The housing 30 encloses a plurality of battery cells 42 that are secured within an interior of the housing by a battery holder 46 (FIG. 15). The housing 30 also encloses battery pack circuitry (not shown). The first portion 34 is symmetrical about the longitudinal axis A, engageable with the charger 18, and includes a wall 50 and a battery pack interface 54 that extends from the wall 50. The battery pack interface 54 includes a terminal block 58 with openings 62 extending therethrough that allow access to battery pack terminals (not shown) positioned within the housing 30. On opposite sides of the battery pack interface 54 are rails 66, 70 and grooves 74, 78 that are defined between the respective rail 66, 70 and the wall 50. A first plurality of vent holes 82 and a second plurality of vent holes 86 extend through the wall 50 and are in communication with the interior of the battery pack 14B. The first plurality of vent holes 82 and the second plurality of vent holes 86 are positioned on opposite sides of the battery pack interface 54 and are adjacent to and parallel to the respective rails 66, 70 and grooves 74, 78. The battery pack 14B further includes a latch mechanism 90 including an actuator 94 and a latch member 98. In the illustrated construction, there are two latch mechanisms 90 on opposite sides of the first portion 34, but in other constructions there may be a single latch mechanism. The second portion 38 includes a plurality of holes 102 that allow communication between the interior of the battery pack 14B and outside of the battery pack 14B.

Each battery pack 14A, 14B is connectable to and operable to power various motorized power tools (e.g., a cut-off

5

saw, a miter saw, a table saw, a core drill, an auger, a breaker, a demolition hammer, a compactor, a vibrator, a compressor, a drain cleaner, a welder, a cable tugger, a pump, etc.), outdoor tools (e.g., a chain saw, a string trimmer, a hedge trimmer, a blower, a lawn mower, etc.), other motorized devices (e.g., vehicles, utility carts, a material handling cart, etc.), and non-motorized electrical devices (e.g., a power supply, a light, an AC/DC adapter, a generator, etc.).

With renewed reference to FIGS. 1-5, the charger 10 has a charger housing 100 including a front wall 110, a rear wall 114, a top wall 118, a bottom wall 122, a first side wall 126, and a second side wall 130. The front wall 114 includes a first portion 134 and a second portion 138 that is inclined relative to the first portion 134. The first and the second side walls 126, 130 are spaced apart from one another, and a transverse axis B of the charger housing 100 is defined therebetween. As shown, the charger housing 100 has a top portion 142 (FIG. 2) and an opposite bottom portion 146 coupled to the top portion 142 (e.g., by fasteners (not shown)). The housing portions 142, 146 may be formed of plastic with each molded as a single piece. In other constructions, the housing 100 may include other housing configurations (e.g., a clamshell housing, etc.). A power input port 150 for connection to a power supply (e.g., through a power cord, not shown) is positioned in the housing 100. In the illustrated construction, the power input port 150 is positioned in the front wall, but in other constructions the power input port 150 may be positioned at any suitable location on the housing.

With reference to FIGS. 7-8, charger electronics 154 are supported by the bottom portion 146 and specifically the bottom wall 122. The charger electronics 154 are operable to output a charging current to one or both of the battery packs 14A, 14B to charge the battery packs 14A, 14B. The charger electronics 154 include, among other things, a printed circuit board (PCB) 158 and a charger microcontroller (not shown). The charger electronics 154 may include a charging circuit portion (not shown; e.g., on separate PCBs) for each of the battery packs 14A, 14B so that each battery pack 14A, 14B may be charged simultaneously and independently. The charging current provided to each battery pack 14A, 14B may be the same or different.

The charger housing 100 further includes charger interfaces 200A, 200B engageable with and able to support the respective battery packs 14A, 14B. Each charger interface 200A, 200B is at least partially positioned substantially on the front wall of the housing (e.g., at least partially on the inclined portion 138).

With reference to FIGS. 1-3, 5 and 9, the charger interface 200A defines an insertion axis C (FIG. 9) and a recess 204 (e.g., a battery pack receiving port) defined by the top wall 118 and the inclined portion 138 of the front wall 110. The insertion axis C is perpendicular to the transverse axis B of the housing and at non-perpendicular and non-parallel angle relative to the inclined portion 138 of the front wall 110. The recess 204 is configured to receive at least a portion (e.g., the tower 208, FIG. 13) of the battery pack 14A. A first set of charger terminals 212 (FIG. 3) extend from within the housing 100 through holes into the recess 204. The charger terminals 212 are configured to electrically connect to battery pack terminals (not shown) of the battery pack 14A received in the recess 204 for charging.

With reference to FIGS. 1-3, 5, and 9-10, the charger interface 200B defines an insertion axis D (FIG. 9) and includes rail members 220, 224 and a charger terminal block 228 (FIG. 1). The rail members 220, 224 are spaced apart, substantially parallel, and positioned on the inclined portion

6

138 of the front wall 110. An aperture 232, 236 extends through each of the rail members 220, 224. The apertures 232, 236 are each in communication with an interior 240 (FIG. 10) of the charger housing 100 via a first chamber 244 positioned within the housing 100. A groove 248, 252 is defined between the inclined portion 138 of the front wall 110 and the associated rail member 220, 224. The insertion axis D is oriented differently than the insertion axis B. In particular, the insertion axis D is parallel to the rail members 220, 224. The rail members 220, 224 and grooves 248, 252 are engageable with respective grooves 74, 78 and rails 66, 70 on the battery pack 14B. The apertures 232, 236 in the rail members 220, 224 of the charger interface 200B are configured to align and communicate with the respective first plurality of vent holes 82 and second plurality of vent holes 86 in the wall 50 of the first portion 34 of the battery pack 14B (FIG. 16). The charger terminal block 228 is positioned within an aperture 256 in the inclined portion 138 positioned between the rail members 220, 224. The charger terminal block 228 includes a second set of charger terminals 260 (FIG. 1) configured to electrically connect to battery pack terminals of the battery pack 14B for charging.

With reference to at least FIG. 6, the first chamber 244 is defined by a first chamber wall 270 that extends from the inclined portion 138 of the front wall 110 into the charger housing 100. As shown in FIG. 6, the first chamber wall 270 defines a polygonal shaped aperture 274 and has posts 278 extending therefrom. The first chamber 244 defines an axis E (FIG. 9) that is perpendicular to the inclined portion 138 of the front wall 110. A slot 282 (FIG. 6) extends through the chamber 270.

As shown in FIGS. 7-11, a first fan 300 or blower (e.g., a DC blower) is positioned adjacent to the first chamber 244. With respect to FIGS. 7, 8, and 11, the first fan 300 includes a fan housing 304 that is positioned adjacent and coupled to the first chamber wall 270. In particular, the fan housing 304 includes through-holes 308 (FIGS. 7 and 11) that receive the posts 278 extending from the chamber wall 270. An opening 312 of the fan housing 304 is aligned with the first chamber 244 (e.g., along the axis E), and one or more blades 316 are rotatably positioned therein. The first fan 300 is a multi-speed fan operable to rotate at more than one speed. The speed at which the fan 300 rotates may be determined based on a temperature of one or more of the charger electronics 154 or the battery pack 14B. In some constructions, at full speed, the first fan 300 generates an air flow of between about 13.6 m³/hour and about 25.5 m³/hour. Still further, in some constructions, the first fan 300 may generate an air flow of about 20.4 cubic feet per minute (CFM) and up to about 35 m³/hour or less. In still other constructions, at full speed, the fan 300 may generate an air flow that is less than or greater than 31 CFM.

An air passage member 350 is positioned within the housing 100 and extends from the first fan 300 through an opening 354 (FIG. 9) in the rear wall 114. With respect to FIGS. 7, 8, 11, and 12, the air passage member 350 includes a hollow body 358 that has a first (e.g., top) wall 362, a second (e.g., bottom) wall 366 opposite the first wall 362, and a pair of side walls 370, 374 that are coupled on opposite sides of the first and second walls 362, 366 and extend therebetween. A first angled wall 378 is coupled to and extends from the first wall 362 and extends between the pair of side walls 370, 374. The first angled wall 378 is coupled to a second angled wall 382. The second angled wall 382 is coupled to and extends from the second wall 366, and extends between the pair of side walls 370, 374. An aperture

386 extends through the first angled wall **378** and is in communication with a second chamber **390** defined by the hollow body **358**.

The air passage member **350**, and therefore the second chamber **390**, includes two axes F, H. One of the axes F passes through a center of the aperture **386** in the first angled wall **378** and is configured to be coincident with the axis E, and the other of the axes H extends between and parallel to the first and second walls **362**, **366**. The F, H are positioned at a non-perpendicular and non-parallel angle relative to one another. In the illustrated construction, the angle between axes F and H is oblique, and measures between 100 degrees and 150 degrees. A preferred angle between the axes F and H is 129 degrees. In other or additional embodiments, the angle between F and H may be perpendicular or acute. Moreover, the axis H is positioned at a non-perpendicular and non-parallel angle relative to the insertion axis D. In the illustrated construction, the angle between axes D and H is acute, and measures between 15 degrees and 60 degrees. A preferred angle between the axes D and H is 39 degrees. In other or additional embodiments, the angle between D and H may be perpendicular or oblique. The air passage may be constructed from plastic acrylonitrile butadiene styrene (ABS), polycarbonate acrylonitrile butadiene styrene (PC-ABS), and/or aluminum, although other suitable thermoplastic, thermoplastic polymers, and metals may also or alternatively be used.

As shown in FIGS. 9-12, the air passage member **350** is coupled to the first fan **300**. In particular, the first angled wall **378** includes through-holes **394** that are aligned with the through-holes **308** in the fan housing **304** and receive the posts **278**. Accordingly, the aperture **386** in the first inclined wall **378** is aligned with the opening **312** in the first fan **300** and the first chamber **244** and allows communication between the first chamber **244** and the second chamber **390**. The axis F of the aperture **386** is coincident with the axis E of the first chamber **240**, as discussed above, and the axis H is parallel to the bottom wall **122** of the housing **100**. Also, the air passage member **350** has a length L that is more than 50% of the depth DI of the housing **100**. Preferably, the length L of the air passage member **350** ranges from 55% of the depth DI of the housing to 75% of the depth DI of the housing **100**.

With reference to FIGS. 9 and 16, the first chamber **244**, the opening **312** in the fan housing **304** and the second chamber **390** define a first air flow path or passage X that receives a first air flow X from outside the charger housing **100**. A distal or protruding end **398** of the air passage member **350** extending from the housing **100** defines an air inlet and the apertures **232**, **236** define the air outlet of the first air passage such that the first air flow X moves from outside the charger housing **100** to the battery pack **14B** sequentially through the second chamber **390**, the fan **300** and the first chamber **244**. Therefore, when actuated, the fan **300** sucks air into the air passage such that the first air flow X travels from outside the charger housing **100** laterally (e.g., horizontally) through the second chamber **390** along the axis H and vertically through the fan **300** and first chamber **244** along the axes E, F. The first air flow X then enters the battery pack **14B** via the apertures **232**, **236** in the rails **220**, **224** of the charger interface **200B** and the first plurality of vent holes **82** and the second plurality of vent holes **86** in the first portion **34** of the battery pack **14B**. Once within the battery pack housing **100** the first air flow X is guided from the respective plurality of vent holes **82**, **86** between the battery cells **42** and the second portion **38** of the battery pack **14B** towards the holes **102** in the second portion

38 of the battery pack **14B**. Accordingly, the first air flow X may be used to cool the battery cells **42** during charging of the first battery pack **14B**.

In other constructions, the distal or protruding end **398** may define the air outlet and the apertures **232**, **236** may define the air inlet of the first air passage such that the first air flow X moves from the battery pack **14B** to outside of the charger housing **100** sequentially through the first and second plurality of vent holes **82**, **86**, the first chamber **244**, the fan **300**, and second chamber **390**. In such case, when actuated, the fan **300** sucks air into the air passage thereby generating a reverse air flow (e.g., airflow that moves in the reverse of the airflow X). Therefore, air travels from inside the battery pack vertically through the first chamber **244** and the fan **300** along the axes E, F and laterally (e.g., horizontally) through the second chamber **390** along the axis H. Accordingly, the first air flow X or the reverse airflow may be used to cool the battery cells **42** during charging of the first battery pack **14B**.

In some constructions, air may be guided in a first direction (e.g., to the battery pack **54**) through one air passage **350** and may be guided in a second direction (e.g., away from the battery pack **54**) through another air passage. Moreover, one of the air passages may include a heating element or a cooling member that is in electrical communication with the charger electronics. A movable valve or seal selectively opens and closes one or both of the air passages. A similar, exemplary view of this construction is shown in FIG. 24C. The heating element or cooling member may be actuated to blow air that is higher (e.g., via the heating element) or lower (e.g., via the cooling element) than ambient air temperature into the battery pack **54** to respectively heat up the cells or cool the cells of the battery pack **54**. The seal is movable between a first, closed position when the heating element or cooling element is not being used and a second, open position when the heating element or cooling element is being used. The heating element or cooling element and the seal may be manually actuatable by a user or automatically actuatable via the charger electronics (e.g., a controller and a temperature sensor). The other of the air passages may always open to suck or blow ambient air into battery pack **54** via charger **10**, as discussed herein. In other constructions, the seal may alternatively open and close the air passages, depending on which is being used.

In other constructions, the air passage **350** may be divided into separate air passages thereby creating a dual-air passage member. In these constructions, air may be guided in a first direction (e.g., to the battery pack **54**) through one of the dual-air passages and may be guided in a second direction (e.g., away from the battery pack **54**) through the other of the dual-air passages. Similarly, one of the dual-air passages may include the heating element or the cooling member that is in electrical communication with the charger electronics, such that air is guided through the dual-air passages, as discussed above and below with respect to FIG. 24C.

In other or additional constructions, the first chamber may be omitted. In such case, the housing may include posts extending therefrom that receive and support the fan **304** and the air passage member **350**.

In still other constructions, the charger interface **200B** may be positioned and supported by other walls of the housing **100**. Similarly, the air passage member **350** may be extended through an opening in another wall (e.g., the bottom wall **122** or one of the side walls **126**, **130**) of the housing **100**. Accordingly, the air passage member **350** may include other suitable configurations such that the air passage member **350** may extend through and from other walls of the

housing 100 to the apertures 232, 236 in the rails 220, 224 of the charger interface 200B.

In still other or additional constructions (FIGS. 17 and 18), the air passage member 350 may be omitted and the aperture in the rear wall 114 replaced by vent holes or slots 400 (e.g., an air inlet). Moreover, there may be additional vent holes 400 in other walls as well, or the vent holes 400 may be placed in other walls of the housing (e.g., closer or further from the fan 300). Therefore, when actuated, the fan 300 sucks air into interior of the charger housing 100 (rather than the first air passage) such that the first air flow X' travels from outside the charger housing 100 laterally through the vent holes 400 to the fan 300 and into the first chamber 244 along the axes E, F. The first air flow X' then enters the battery pack 14B as discussed above with respect to the first airflow X. Alternatively, when actuated, the fan 300 sucks air into interior of the charger housing 100 (rather than the first air passage) such that the reverse air flow (e.g., airflow that moves in the reverse of airflow X') travels from the battery pack 14B into the outside the charger housing 100 into the first chamber 244 along the axes E, F to the fan 300, and laterally from the fan 300 through the vent holes 400.

With reference to FIGS. 1-4, the charger 10 may have other or auxiliary air passages that extend between other an air inlets and outlets in the housing 100. For example, the housing 100 also defines air inlets 520, 524 (FIG. 2) positioned in the side wall 126 of the housing 100 and proximate the rear wall 114. The inlets 520, 524 include slots 528, 532 extending through the side wall 126. In other constructions (not shown), the slots 520, 524 may extend in a latitudinal direction, a combination longitudinal/latitudinal direction, etc. The slots 528, 532 are configured to facilitate air flow entering the housing 100.

The housing 100 further includes an air outlet 500 positioned below the first charging interface 200A (e.g., the recess 204). As such, the illustrated air outlet 500 is below the battery pack 14A when coupled to the charger 10. In addition, the illustrated outlet 500 includes slots 504 (e.g., longitudinal slots) defined in and extending through the inclined portion 138 and, partially, by the first portion 132. In other constructions (not shown), the slots 504 may extend in a latitudinal direction, a combination longitudinal/latitudinal direction, etc. The slots 504 are configured to facilitate air flow into the housing 100. Other air outlet 508 configured as one or more slots 512 may be extend through the bottom wall 122 as well (FIG. 4). The inlets 520, 524 and the outlets 500, 508 are positioned on different locations of the housing 100.

With reference to FIGS. 7 and 8, the charger 10 may also include a heat sink 530 and a second fan 534 or blower (e.g., a DC blower) within the housing 100 to provide a heat dissipating structure. A temperature sensor (not shown) is disposed in the housing 100 and positioned near the charger electronics 154 (e.g., near the component(s) generating the most heat (e.g., the CPU, the transformer, field effect transistors (FETs), etc.)) or the heat sink 530. In the illustrated construction, the heat sink 530 is disposed in the housing 100 proximate the rear wall 114. In other constructions (not shown), the heat sink 530 may be positioned at other locations in the housing 100. The heat sink 530 is in heat transfer relationship with components of the charger electronics 154 (e.g., is mounted onto and in contact with the PCB 158). In other words, heat transfers from the heat-generating components of the charger 10 to the heat sink 530 through conduction.

In the illustrated construction, the heat sink 530 is formed of heat-conducting material, such as, for example, alumi-

num, and extends between opposite ends 538, 542. Furthermore, the illustrated heat sink 530 is constructed of one or more hollow tubes 546 (three are shown in FIG. 7), each having a rectangular shape and stacked above one another. The tubes 546 extend between the opposite ends 538, 542. As such, the illustrated heat sink 530 forms a tubular heat sink. The first end 542 forms an inlet of each tube 546 for air flow to enter the heat sink 530, and the second end 538 forms an outlet of each tube 546 for air flow to exit the heat sink 530. As shown in FIG. 7, the inlet of each tube 546 is angled toward the front of the housing 100. In other or additional embodiments, the heat sinks may be flat plates rather than tubular heat sinks.

The second fan 534 is positioned between the first end 542 of the heat sink 530 and the inlets 520, 524. A baffle 550 extends between the first end 542 and the fan 534 for directing air flow from the heat sink 530 to the outlets 500, 508. The second fan 534 is a multi-speed fan operable to rotate at more than one speed and directs air flow from the inlets 520, 524 through the housing 100 and to the outlets 500, 508. The speed at which the fan 534 rotates may be determined based on a temperature of one or more of the charger electronics 154, the heat sink 530, a supported battery pack 14A, 14B, etc. The temperature sensor is configured to measure the temperature and transmit a signal output to the microcontroller for determining the temperature of the charger 10. Subsequently, the microcontroller controls the speed of the second fan 354 based on the temperature (e.g., of the heat sink 530, as illustrated). In some constructions, at full speed, the second fan 354 generates an air flow of between about 13.6 m³/hour and about 25.5 m³/hour. Still further, in some constructions, the second fan 154 may generate an air flow of about 20.4 cubic feet per minute (CFM) and up to about 35 m³/hour or less.

As shown in FIG. 7, the charger 10 defines a second flow path or passage Y through the housing 100 that is separate from the first airflow X, X'. In the illustrated construction, second air flow flows along the flow path Y from the inlets 520, 524, over the charger electronics 154 (e.g., the PCB 158) to the inlet of the heat sink 530, and through the heat sink 530 to the outlets 500, 508. The second fan 354 directs the second air flow along the flow path Y. Furthermore, the second fan 354 directs the second air flow Y into the inlet and outlet 542, 538 of each tube 546. The second air flow Y operates to dissipate heat generated by the charger electronics 154 from the housing 100. In other constructions (not shown), the second fan 354 may be operated in reverse such that the second flow path Y through the housing 100 is reversed.

In operation, one or both of the battery packs 14A, 14B are coupled to the respective charger interface 200A, 200B (e.g., the supporting sections 64A, 64B) for charging. The first set of terminals 212 electrically connect with the battery pack terminals of the first battery 14A, and/or the second set of terminals 260 electrically connect with the battery terminals of the second battery 14B. The charger 10 supplies charging current to the first and/or second battery 14A, 14B.

FIGS. 19-31 illustrate a charging system according to another construction including a battery charger 600 operable to charge a battery pack 614. The battery pack of FIGS. 19-29 is similar to the battery pack of FIGS. 14A-15. Therefore, the battery pack 614 will not be described in detail and like reference numerals plus "600" will refer to like structure in the description below.

With respect to FIG. 19, the battery charger 600 includes a housing 750, 754 having a longitudinal axis J and a frame member 758 coupled to the housing. The housing 750, 754

includes a battery pack receiving portion **750** and a charger electronics portion **754** that is coupled to the battery pack receiving portion **750**. In some constructions, the battery pack receiving portion **750** may be sealed with respect to the charger electronics portions **754**.

As shown in FIG. **21**, the battery pack receiving portion **750** includes a first wall **770**, a second wall **774**, a third wall **778**, a fourth wall **782**, a fifth wall **786**, and a sixth wall **790**. In the illustrated construction the first wall **770** is the top wall and the second wall **774** is the bottom wall and therefore the first wall **770** and the second wall **774** are spaced apart from one another. Specifically, the first wall **770** and the second wall **774** are positioned opposite one another. The third wall **778** is the front wall and the fourth wall **782** is the rear wall and therefore the third and fourth walls **778**, **782** are spaced apart from one another. Specifically, the third wall **778** and the fourth wall **782** are positioned opposite one another. The fifth and sixth walls **786**, **790** couple the first wall **770**, the second wall **774**, third wall **778** and fourth wall **782**. In the illustrated embodiment and as shown in FIG. **21**, the battery pack receiving portion **750** is defined by a first portion **794** and a second portion **798**. The first portion **794** includes the first wall **770** and the fourth wall **782**, while the second portion **798** includes the second wall **774**. The first portion **794** and the second portion **798** together define the third wall **778**, the fifth wall **786**, and the sixth wall **790**. The first and second portions **794**, **798**, and therefore the walls **770**, **774**, **778**, **782**, **786**, **790**, are molded from a plastic material.

The first wall **770** includes an elongate opening **806** and a plurality of vent holes **810** extending therethrough. The elongate opening **806** has a closed end **814** adjacent the fourth wall **782** and an open end **818** adjacent the third wall **778**. A charger interface **822** is positioned in the elongate opening **806**, at least partially supported by the first wall **770**, and configured to receive the battery pack interface **654**. A first set of the plurality of vent holes **810** are positioned on one side of the elongate opening **806** and a second set of the plurality of vent holes **810** are positioned on an opposite side of the elongate opening **806**. The second wall **774** includes a plurality of vent holes **826** (FIGS. **22** and **23**) extending therethrough. A handle **830** extends from and is movable (e.g., pivotable) relative to the third wall **778**.

The charger interface **822** of FIGS. **19-21** has an insertion axis **K** (FIG. **21**) that is parallel to the longitudinal axis **J** and is at least partially defined by a first supporting member **840** and a second supporting member **844**. With respect to FIG. **22**, the first supporting member **840** includes a first side wall **848** positioned adjacent one side of the elongate opening **806**, a first recessed wall **852** extending from the first side wall **848**, and a first portion **856** of a terminal block **860** positioned within the battery pack receiving portion **750** adjacent the closed end **814**. The first side wall **848** includes a first rail member **864** extending therefrom and a first groove **868** is positioned between the first rail member **864** and the first recessed wall **852**. The second supporting member **844** includes a second side wall **876** positioned adjacent an opposite side of the elongate opening **806**, a second recessed wall **880** extending from the second side wall **876**, and a second portion **884** of the terminal block **860** positioned within the battery pack receiving portion **750** adjacent the closed end **814** and the first portion **856** of the terminal block **860**. The second side wall **876** includes a second rail member **888** extending therefrom and a second groove **892** is positioned between the second rail member **888** and the second recessed wall **880**. A surface of each of the rail members **864**, **888** is coextensive with an adjacent

surface of the first wall **770** and the surfaces of the recessed walls **852**, **880** are recessed relative to the surface of the first wall **770**. The rail members **864**, **888** are spaced apart and substantially parallel to one another. The insertion axis **K** is parallel to the rail members **864**, **888**. Charging terminals **896** extend from each portion **856**, **884** of the terminal block **860**.

As shown in FIGS. **21-22**, the first supporting member **840** further includes one or more (in the illustrated embodiment two) fan or blower support members **910** extending from the first side wall **848** opposite the first recess wall **852**. The fan supporting members **910** of the first supporting member **840** are positioned within the battery pack receiving portion **750** adjacent the first set of vent holes **810**. Similarly, the second supporting member **844** further includes one or more (in the illustrated embodiment two) fan or blower support members **914** extending from the second side wall **876** opposite the second recess wall **880**. The fan supporting members **914** of the second supporting member **844** are positioned within the battery pack receiving portion **750** adjacent the second set of vent holes **810**.

As shown in FIGS. **22-23**, each of the fan supporting members **910**, **914** receives and supports a fan **918** or blower (e.g., a DC blower) such that a plurality of fans **918** or blowers is positioned within the battery pack receiving portion **750** between the first wall **770** and the second wall **774**. Each of the fans **918** is therefore positioned at or adjacent to one or more of the vent holes **810** in the first wall **770**. The fans **918** by the fan supporting members **910** are supported such that they are spaced apart from (e.g., positioned above) the second wall **774** and one or more vent holes **826** in the second wall **774**. In other constructions, there may additionally or alternatively be one or more vent holes in one or both of the fifth and sixth side walls **786**, **790** (FIG. **24b**). In the illustrated construction, each of the fans **918** has an axis of rotation **L** that is positioned at a non-parallel angle relative to the insertion axis **K**. In the illustrated construction, each of the fans **918** has an axis of rotation **L** that is perpendicular to the insertion axis **K**. In the illustrated construction the axes of rotation **L** of the fans **918** are oriented in parallel to one another, but in other or additional constructions, the axes of rotation **K** may be oriented in parallel or at a non-parallel angles relative to one another.

In the constructions of FIGS. **19-27**, a cooling air path or passage **922** extends from one or more of the plurality of vent holes **810** in the first wall **770** to one or more of the plurality of vent holes **826** in the second wall **774** through each of the fans **918**. Accordingly, each of the fans **918** is operable to move cooling air between one or more of the vent holes **810** in the first wall **770** and one or more vent holes **826** in the second wall **774**. In the illustrated embodiment, there are four fans **918**, two on each side of the charger interface **822**. Accordingly, there are four cooling air passages **922** defined within and extending through the battery pack receiving portion **750** of the charger housing. In other or additional constructions, there may be other or additional fans **918** and cooling air passages **922** defined within and extending through the battery pack receiving portion **750**. Additionally, as shown in FIG. **26**, the cooling air passages **922** guide airflow in a direction parallel to the axis of rotation **L** of the respective fan **918**. Therefore, in the illustrated construction, the cooling air of the cooling air passages **922** moves perpendicular to the axis of the insertion **K**. Also, as shown in FIG. **27**, air moves vertically. In other or additional constructions, there may be vent holes elsewhere in the battery pack receiving portion such that the

13

cooling air passages may move air in a direction perpendicular and vertically relative to the insertion axis K but also in a non-perpendicular direction and horizontally relative to the insertion axis K.

In other or additional constructions, an air passage member, similar to that described above with respect to FIGS. 1-18, may define each of the cooling air passage 922. That is, in other or additional constructions, the cooling air passage 922 may be at least partially defined by an air passage member having a hollow body having a first end positioned at or adjacent the fan support member and the second end positioned at or adjacent to the one or more vent holes in the second wall 774. In these constructions, each fan 918 would be supported at or adjacent the first end of the air passage member.

Moreover, as shown in FIGS. 24A, 24B, although only one is shown, a duct member 928 may be positioned on each side of the charger interface 822. The duct members 928 may each define one or more of air passage members 932, 936 through which a respective cooling air passage 922 flows. In other words, the duct member 928 may define a first air passage member 932 and a second air passage member 936. The respective first air passage member 932 includes an inlet 940 that is configured to be positioned at or adjacent to and in communication with the respective one of the fans 918 and the air passage member 936 includes an inlet 944 that is configured to be positioned at or adjacent to and in communication with an adjacent fan 918. Each of the respective first and second air passage members 932, 936 includes an outlet 948, 952 that is at or adjacent to and in communication with one or more of the second plurality of vent holes 826.

As shown in FIG. 24C, air may be guided in a first direction (e.g., to the battery pack 654) through one of the air passages 932 and may be guided in a second direction (e.g., away from the battery pack 654) through the other of the air passages 936. Moreover, one of the air passages 932, 936 may include a heating element or a cooling member 934 that is in electrical communication with the charger electronics. A movable valve or seal 935 selectively opens and closes the air passage 932a, 932b. In the illustrated construction, the seal 935 is positioned between the fan 918 and the vent holes 810, but in other constructions, the seal 935 may be positioned elsewhere (e.g., within the air passage 932, 936). The heating element or cooling member 934 may be actuated to blow air that is higher (e.g., via the heating element) or lower (e.g., via the cooling element) than ambient air temperature into the battery pack 654 to respectively heat up the cells or cool the cells of the battery pack 654. The seal 935 is movable between a first, closed position when the heating element or cooling element 934 is not being used and a second, open position when the heating element or cooling element 934 is being used. The heating element or cooling element 934 and the seal 935 may be manually actuatable by a user or automatically actuatable via the charger electronics (e.g., a controller and a temperature sensor). The other of the air passages 932, 936 may always open to suck or blow ambient air into battery pack 654 via charger 600, as discussed herein. In other constructions, the seal 935 may alternatively open and close the air passages 932, 936, depending on which is being used.

In other constructions, one or both of the air passage members 932, 936 may be divided into separate air passages thereby creating a dual-air passage member. In these constructions, air may be guided in a first direction (e.g., to the battery pack 654) through one of the dual-air passages and may be guided in a second direction (e.g., away from the

14

battery pack 654) through the other of the dual-air passages 936. Similarly, one of the dual-air passages may include the heating element or the cooling member 934 that is in electrical communication with the charger electronics, such that air is guided through the dual-air passages as discussed above with respect to FIG. 24C.

As shown in FIG. 25, the charger 600 includes a drain member 960 configured to drain fluid from the charger interface 822 to an area outside of the charger 600. In the illustrated construction, the drain member 960 includes a first end 964 is positioned adjacent to and in communication with an area adjacent the terminal block 860 and second end 968 positioned at or adjacent to and in communication with an aperture (not shown) that extends through one of the other walls of the battery pack receiving portion 750. In the illustrated construction, the second end 968 is positioned at or adjacent to and in communication with an aperture that extends through the second wall 774. In other constructions, the second end 968 is positioned at or adjacent to and in communication with an aperture that extends through the another of the walls that is spaced apart from the first wall 770.

With respect to FIGS. 22, 23, 26 and 27, the charger electronics portion 754 includes a cover 1000, a fan 1004 or blower (e.g., a DC blower), and a charger electronics housing 1008. In the construction of FIG. 24, the cover 1000 is formed from a plastic material and includes a rear wall 1012 (FIG. 27), a top wall 1016 (FIGS. 22, 23, 27), a bottom wall 1020 (FIG. 27), a first side wall 1024 (FIG. 23), and a second side wall 1028 (FIG. 22). Each of the walls 1016, 1020, 1024, 1028 is coupled to and extends from the rear wall 1012. With respect to FIG. 27, the rear wall 1012 includes a first plurality of vent holes 1032 and one or more of the walls 1016, 1020, 1024, 1028 includes a second plurality of vent holes 1036. In the illustrated construction, the second plurality of vent holes 1036 are in the bottom wall 1020, but in other constructions the second plurality of vent holes 1036 could be in the other walls, as well or alternatively.

With respect to FIG. 26, the charger electronics housing 1008 is constructed of metal and has a plurality of walls that define a sealed (e.g., fluid and debris) box-like structure that encloses the charger electronics 1050. In the illustrated construction, the metal is aluminum, but other suitable metals may be used instead or in addition to aluminum. A first wall 1054 of the charger electronics housing 1008 includes a plurality of fins 1058 (or other heat dissipating members) extending therefrom and a second, opposite wall 1062 supports the charger electronics 1050. A plurality side walls 1062 extend from the first wall 1054 toward the second wall 1058. The plurality of side walls 1062 is coupled to the second wall to sealably (e.g., to enclose the charger electronics therein). The plurality of fins 1058 forms a recess 1066 that is sized and shaped to receive the fan 1004. The charger electronics 1050 include, among other things, a printed circuit board (PCB) 1070 and a charger microcontroller. The charger electronics may include any suitable configuration. In the embodiment of FIG. 26, the PCB 170 is positioned adjacent the first wall 1054.

The cover 1000 is coupled to (e.g., by fasteners or the like, not shown) to the battery pack receiving portion 750. Specifically, the walls of the cover 1000 are coupled to the second, fourth, fifth, and sixth walls 774, 782, 786, 790 of the battery pack receiving portion 750. The fan 1004 and the charger electronics housing 1008 are positioned between and enclosed by the cover 1000 and the battery pack receiving portion 750. As shown the first wall 1054 having the fins 1054 is positioned adjacent the rear wall 1012 such

15

that the fan **1004**, which is received in and movable (e.g., rotatable) relative to the recess **1066**, is positioned adjacent the first plurality of vent holes **1032**. A cooling air passage **1080** is defined between the first plurality of vent holes **1032** and the second plurality of vent holes **1036** and the fan **1004** is positioned between the first plurality of vent holes **1032** and the second plurality of vent holes **1036**.

In use and as shown in FIGS. **19** and **26**, the rail members **864**, **888**, and grooves **868**, **892** are engageable with respective grooves **674**, **678** and rails **666**, **670** on the battery pack **614** to bring the charging terminals **896** into electrical and physical engagement with the battery terminals such that the charger **600** is able to charge the batteries of the battery pack **614**. When the battery pack **614** is coupled to the charger interface **822**, the vent holes **810** in the first wall **770** are configured to align and communicate with the respective first plurality of vent holes **682** and second plurality of vent holes **686** in the wall **650** of the first portion **634** of the battery pack **614**. When the battery pack **614** is coupled to the battery charger **600**, each of the fans **918** of the battery pack receiving portion **750** are operable to move cooling air between an interior of the battery pack **614** and an area outside of the charger **600** via the respective cooling air passage **922**. In the illustrated construction, the fans **918** are operable to suck ambient air through holes **702** in the battery pack **614**, through the interior of the battery pack **614**, and into the charger **600** (e.g., the battery pack receiving portion **750**) via the respective plurality of vent holes **810** in the first wall **770** and expel the air through the respective plurality of vent holes **826** in the second wall **774** such that air can move through the battery pack **614** and be removed therefrom by flowing through the charger **600**. In other constructions, the fans **918** suck hot air from within the interior of the battery pack **614** into the charger **600** (e.g., the battery pack receiving portion **750**) via the respective plurality of vent holes **810** in the first wall **770** and expel the air through the respective plurality of vent holes **826** in the second wall **774** such that hot air from within the battery pack **614** is removed therefrom by flowing through the charger **600**. In still other constructions, the fans **918** suck air from an area outside of the charger **600** into the charger **600** (e.g., the battery pack receiving portion **750**) via the respective plurality of vent holes **826** in the second wall **774** and expel the air through the respective plurality of vent holes **810** in the first wall **770** into the vent holes **682**, **686** of the battery pack **614** such that ambient air from the area outside the charger **600** is blown into the battery pack **614** to cool the cells of the battery pack **614**. In other constructions, air having a temperature that is less than or greater than an ambient air temperature may be sucked or blown into the charger **600** and/or battery pack **654**, as discussed above with respect to FIG. **24C**. In some constructions, the battery pack **654** completely covers or extends beyond the first wall **770** such that air is forced through the battery pack **654** and not around the battery pack **654**. The terminal block **860** is not sealed relative to battery pack receiving portion **750** so water or other fluids may drain from the interior of the battery pack **654** and the charger interface **822** to the area outside of the charger **600** via the drain member **960**.

Additionally, the fan **1004** and cooling air passage **1080** of the charger electronics portion **754** are operable to cool the charger electronics housing **1008**. In the illustrated construction, the fan **1004** suck ambient air from an area adjacent the first plurality of vent holes **1032** into the charger electronics portion **754** via the respective first plurality of vent holes **1032** in the rear wall **1012** and expel the air through the respective second plurality of vent holes **1036** in

16

the second wall **1020** such that cooling air moves by convection around (and therefore in any direction within) the charger electronics portion **754** to the second plurality of vent holes **1036**, where the air exits the charger electronics portion **754**. The cooling air passages **922** of the battery pack receiving portion **750** and the cooling air passage **1080** of the charger electronics portion **754** are substantially separate from one another and therefore the air does not mix.

In another construction, illustrated in FIGS. **28** and **30-31**, the charger electronics portion **754** has alternative configuration. The charger electronics portion **754** of FIG. **28** is similar to the charger electronics portion **754** of FIG. **26**, and therefore only the differences between the charger electronics portions **754** of FIG. **26** and FIG. **28** will be discussed.

As shown in FIG. **31**, the first side wall **1024** of the cover **1000** includes the first plurality of vent holes **1032** and the second side wall **1028** of the cover **1000** includes a second plurality of vent holes **1036**. In the construction of FIG. **28**, an air passage member **1100** is integrally formed with or otherwise coupled to one of the plurality of side walls **1062** of the charger electronics housing **1008** and has an air channel **1104** extending therethrough. Accordingly, the side wall **1062** to which the air passage member **1100** is coupled defines a partitioning wall **1062** that separates the interior of the charger electronics housing **1008** (which houses the charger electronics **1050** and is filled with a potting compound) from the air channel **1104** of the air passage member **1100**. The air passage member **1100** has a first open end **1108** and a second open end **1112**. The fan **1004** may be supported at either the first or second open ends **1108**, **1112** of the air passage member **1100**. In the illustrated embodiment, the fan is supported at the first open end **1108**. The air passage member of FIG. **27** has a plurality of fins **1116** (or other heat dissipating members) that extend from one side across at least a portion of the air channel **1104**. In the illustrated construction of FIG. **28**, the air passage member **1100** is not completely enclosed as a side opposite the side wall defines an elongated opening. In other constructions, the side opposite the side wall may be defined a wall that may or may not have openings extending therethrough.

The charger electronics housing of FIG. **28** is positioned within the cover **1000** and between the cover **10000** and battery pack receiving portion **750** such that the first end **1108** of the air passage member **1100**, and therefore, the fan **1004** is positioned adjacent the first or second plurality of vent holes **1032**, **1036** and the second end **1112** of the air passage member **110** is positioned adjacent the other of the first and second plurality of vent holes **1032**, **1036**. A cooling air passage **1140** is defined between the first or second plurality of vent holes **1036**, **1036** in the cover **1000**, the fan **1004**, the second end of the air passage member **1112**, and the other of the first or second plurality of vent holes **1032**, **1036**.

In another construction, illustrated in FIG. **29-31**, the charger electronics portion **754** has alternative configuration. The charger electronics portion **754** of FIG. **29** is similar to the charger electronics portion **754** of FIGS. **26** and **28**, and therefore only the differences between the charger electronics portions of FIGS. **26**, **28** and FIG. **29** will be discussed.

As shown in FIG. **31** and like the cover of FIG. **28**, the first side wall **1024** of the cover **1000** includes the first plurality of vent holes **1032** and the second side wall **1028** of the cover **1000** includes a second plurality of vent holes **1036**. In the construction of FIG. **29**, the charger electronics housing defines a first housing portion **1150** and a second housing portion **1154**. The first housing portion **1150**

17

includes the first wall **1054** and a portion of first and second side walls **1062** extending therefrom. The first and second side walls **1062** are on opposite sides of the housing **1008** from one another. Each of the first and second side walls **1062** defines a flange **1160** extending along the length of the respective wall **1062**. Additionally, a partition wall **1162** extends from the first wall **1054** between the first and second side walls **1062** and extends the length of the first and second side walls **1062**. The partition wall **1162** of the first housing portion **1150** has a flange **1160** projecting therefrom and extending along the length of the wall **1162**. The second housing portion **1154** includes the second wall **1058** and a portion of the first and second side walls **1062** extending therefrom. Each of the first and second side walls **1062** of the second housing portion **1154** have a coupling groove **1164** projecting therefrom and extending along the length of the respective wall **1062**. Additionally, a coupling groove **1164** extends from the second wall **1058** between the first and second side walls **1062** and extends the length of the first and second side walls **1062**. The coupling groove **1164** on each of the first and second side walls **1062** of the second housing portion **1154** is configured to receive the flange **1160** on the respective first and second side wall **1062** of the first housing portion **1150**. Similarly, the coupling groove **1164** extending from the first wall **1054** is configured to receive the flange **1160** of the partition wall **1162**. Each of the first and second side walls **1062** include fins **1172** (or other heat dissipating members) extending therefrom.

A third side wall **1062** of the box-like structure of the charger electronics housing **1008** is coupled (e.g., by fasteners and the like) between the first and second walls **1054**, **1058** and the first and second side walls **1054**, **1058**. The third side wall **1062** defines an aperture extending **1176** therethrough. A fourth side wall **1062** of the box-like structure of the charger electronics housing **1008** is coupled (e.g., by fasteners and the like) between the first and second walls **1054**, **1058** and the first and second side walls **1062**. The fourth side wall **1062** includes a plurality of vent holes **1180** extending therethrough. The partition wall **1162** extends between the third and fourth side walls **1062** of the charger electronics housing **1000**.

An interior of the charge electronics housing **1008** is divided into a first portion **1190** and a second portion **1194** by the partition wall **1162**. The partition wall **1162** ensures that the first portion **1190** and the second portion **1194** are separated from one another. The first portion **1190** is completely enclosed by the partition wall **1162** and the other walls **1054**, **1058**, **1062** of the box-like structure of the charger electronics housing **1008**. The charger electronics **1050** are positioned in the first portion **1190** and a potting compound fills the first portion. In the construction of FIG. **29**, an auxiliary plate **1198** (e.g., aluminum or other metal plate) may support at least a portion of the charger electronics **1050** and be positioned adjacent to and in thermal communication with the partition wall **1162**.

The second portion **1194** is enclosed by the partition wall **1162** and the other walls **1054**, **1058**, **1062** of the box-like structure of the charger electronics housing **1008**. The aperture **1176** of the third side wall **1062** is positioned to be in communication and aligned with the second portion **1194** and accommodates the fan **1004**, which is positioned within the second portion **1194**. The plurality of vent holes **1180** of the fourth side wall **1062** is positioned to be in communication with the second portion **1194** such that the plurality of vent holes **1180** is positioned on the opposite side of the charger electronics housing **1008** from the fan **1004**. In the

18

illustrated construction, the partition wall **1162** includes fins **1202** that extend therefrom into the second portion **1194**.

The charger electronics housing **1008** is positioned within the cover **1000** and between the cover **1000** and battery pack receiving portion **750** such that the fan **1004** is positioned adjacent the first or second plurality of vent holes **1032**, **1036** and the plurality of vent holes **1180** in the charger electronics housing **1008** is positioned adjacent the other of the first and second plurality of vent holes **1032**, **1036**. A cooling air passage **1206** is defined between the first or second plurality of vent holes **1032**, **1036** in the cover **1000**, the fan **1004**, the plurality of vent holes **1180** in the charger electronics housing **1008**, and the other of the first or second plurality of vent holes **1032**, **1036**.

Regardless of the construction of the charger electronics portion **754**, airflow through the battery pack receiving portion **750** remains the same, as discussed with respect to FIG. **26** above. However, as shown in FIG. **31**, using the charger electronics portions **754** of either of FIG. **28** or **29** allows cooling air to flow between opposite sides thereof. In particular, the fan **1004** and cooling air passage **1140**, **1206** of the charger electronics portions **754** of both FIGS. **28** and **29** are operable to cool the charger electronics housing **1008** and dissipate heat therefrom. In the illustrated construction, each of the fans **1004** suck ambient air from an area adjacent the first or second plurality of vent holes **1032**, **1036** (depending on which the fan **1004** is positioned at or adjacent to) into the charger electronics portion **754** via the respective first or second plurality of vent holes **1032**, **1036** and expel the air through the opposite first or second plurality of vent holes **1032**, **1036**. With respect to FIG. **28** specifically, the fan **1004** sucks ambient air from an area adjacent the first or second plurality of vent holes **1032**, **1036** (depending on which the fan **1004** is positioned at or adjacent to) into the charger electronics portion **754** via the respective first or second plurality of vent holes **1032**, **1036** (depending on which the second end **1112** of the air channel **1104** is positioned at or adjacent to) and expel the air through the opposite first or second plurality of vent holes **1032**, **1036** such that cooling air moves through the air channel **1104**. With respect to FIG. **29** specifically, the fan **1004** sucks ambient air from an area adjacent the first or second plurality of vent holes **1032**, **1036** (depending on which the fan **1004** is positioned at or adjacent to) into the charger electronics portion **754** via the respective first or second plurality of vent holes **1032**, **1036** (depending on which plurality of vent holes **1180** in the charger housing **1008** is positioned at or adjacent to) and expel the air through the opposite first or second plurality of vent holes **1032**, **1036** such that cooling air moves through the second portion **1194** of the charger electronics housing **1008**. Regardless, in each of the embodiments of FIGS. **28** and **29**, the cooling air passage guides cooling air along an axis M, which is oriented at a non-parallel angle relative to the insertion axis K. In the illustrated constructions, the axis M is oriented at a perpendicular angle to the insertion axis K.

In the illustrated constructions of FIGS. **19-31**, each of the fans **918**, **1004** is a multi-speed fan operable to rotate at more than one speed. The speed at which each of the fans **918**, **1004** rotates may be determined based on a temperature of one or more of the charger electronics **1050** or the battery pack **614**. In some constructions, at full speed, the fans **918**, **1004** generate an air flow of between about 13.6 m³/hour and about 25.5 m³/hour. Still further, in some constructions, the fans **918**, **1004** may generate an air flow of about 20.4 cubic feet per minute (CFM) and up to about 35 m³/hour or less. In still other constructions, at full speed, the fans **918**,

19

1004 may generate an air flow that is less than or greater than 31 CFM. For example, the fans **918**, **1004** may use 31.64 CFM/amp on each cell in an unloaded condition.

The constructions discussed herein are configured to move air through the respective charger and the battery pack. Other constructions may be configured to move other cooling fluids through the respective charger and battery pack.

Although the disclosure has been described with reference to certain preferred aspects, variations and modifications exist within the scope and spirit of one or more independent aspects of the disclosure as described. Various features and advantages of the disclosure are set forth in the following claims.

What is claimed is:

1. A charger comprising:
 - a housing;
 - an interface positioned on the housing and configured to engage a battery pack, the interface being in communication with an interior of the housing;
 - a fan positioned within the housing; and
 - an air passage member including a first end coupled to the fan and a second end spaced apart from the first end, the second end being in communication with outside of the housing through a wall of the housing, and
 - wherein the fan is operable to suck an air flow from outside the housing and through the air passage member to the battery pack interface.
2. The charger of claim 1, wherein the second end of the air passage member is in communication with outside of the housing through a wall of the housing opposite the interface.
3. The charger of claim 2, wherein
 - the housing includes a front wall, a rear wall, and a depth between the front wall and the rear wall, and
 - the air passage member has a length that is at least 50% of the depth.
4. The charger of claim 1, wherein the air passage member includes a first axis and a second axis, the second axis being oriented at an oblique angle relative to the first axis.
5. The charger of claim 1, wherein
 - the interface further includes a first rail and a second rail, and
 - each of the first rail and the second rail includes an aperture extending therethrough.
6. A charger comprising:
 - a housing;
 - charger electronics positioned within the housing;
 - an interface positioned on the housing and configured to engage a battery pack, the interface including charging terminals in electrical communication with the charger electronics;
 - a first fan positioned in the housing and in communication with the interface;
 - an air passage member including
 - a first end positioned within the housing and in communication with the first fan, and
 - a second end spaced apart from the first end, the second end protruding through a wall of the housing, the

20

second end in communication with the first end and with an outside of the housing; and

a second fan positioned in the housing and operable to cool the charger electronics.

7. The charger of claim 6, wherein the first fan is positioned closer to the interface than the second fan is to the interface.

8. The charger of claim 6, wherein

- the first fan creates a first air flow,
- the second fan creates a second air flow, and
- the first air flow and the second air flow are separate from each other.

9. The charger of claim 6, wherein the interface includes two rails, and each of the two rails includes an aperture that is in communication with the air passage member.

10. The charger of claim 6, wherein the housing further includes a heat sink that is in a heat transfer relationship with components of the charger electronics.

11. The charger of claim 6, wherein the first fan is adjacent to a front of the housing.

12. The charger of claim 6, further comprising a second interface positioned on the housing, the second interface configured to engage a second battery pack, the second interface including charging terminals that are in electrical communication with the charger electronics.

13. A charger comprising:

- a housing;
- charger electronics positioned within the housing;
- an interface positioned on the housing and configured to engage a battery pack, the interface including an air flow outlet defined therein;
- an air flow inlet coupled to the housing and protruding beyond a wall of the housing;
- a first fan positioned in the housing, the first fan configured to create an air flow entering through the air flow inlet and exiting through the air flow outlet; and
- a second fan positioned within the housing, the second fan configured to create a second air flow passing over the charger electronics.

14. The charger of claim 13, further comprising an air passage member including a first end positioned within the housing, the first end adjacent the first fan.

15. The charger of claim 13, wherein the air flow and the second air flow are separate from each other.

16. The charger of claim 15, further comprising a heat sink disposed in the housing and in a heat transfer relationship with the charger electronics.

17. The charger of claim 16, wherein the heat sink includes a passage defined therein, and the second air flow travels through the passage.

18. The charger of claim 17, wherein the second fan is positioned adjacent an end of the heat sink.

19. The charger of claim 14, wherein the air passage member further includes a second end spaced apart from the first end, the second end including the air flow inlet defined therein.

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